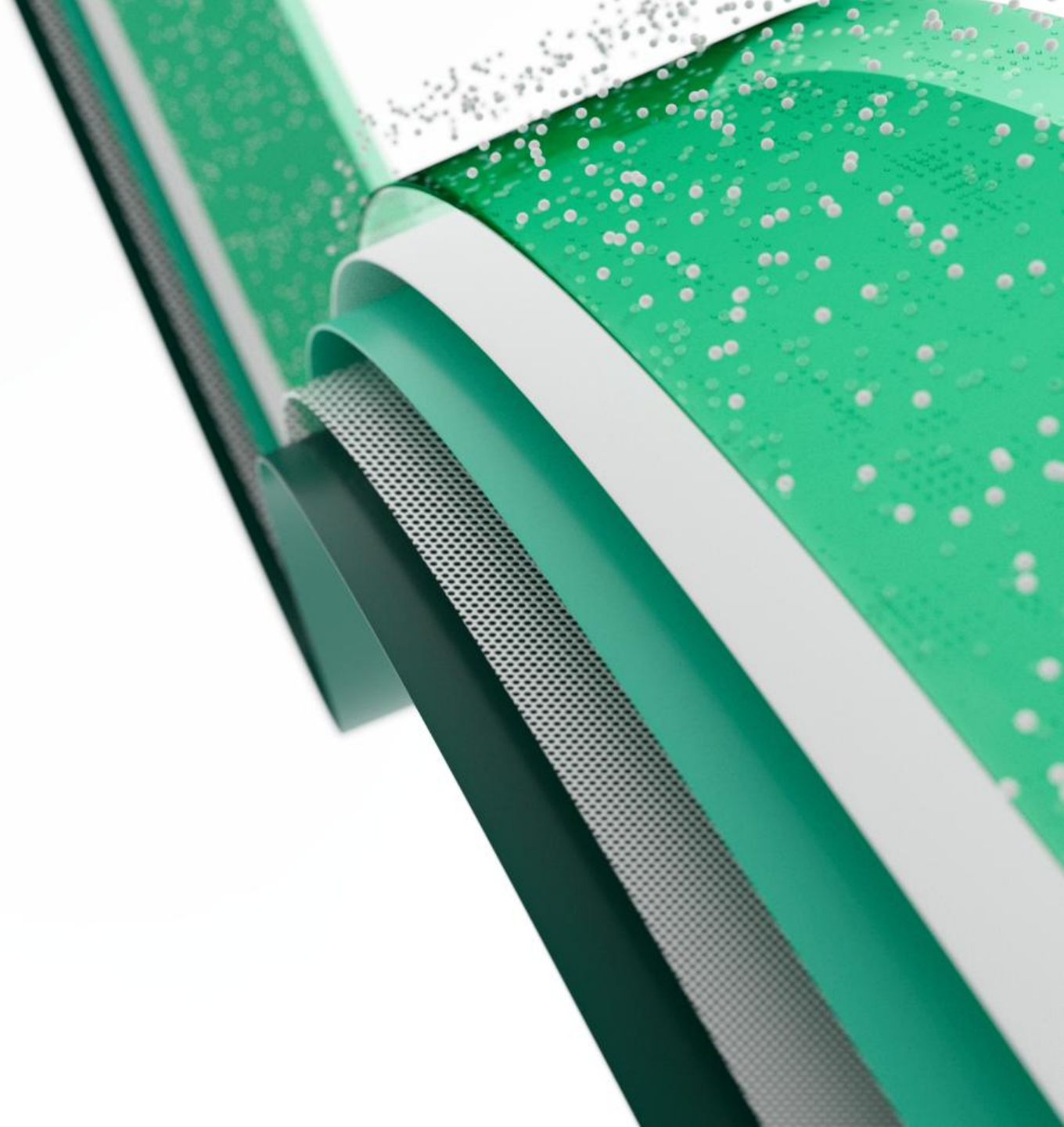


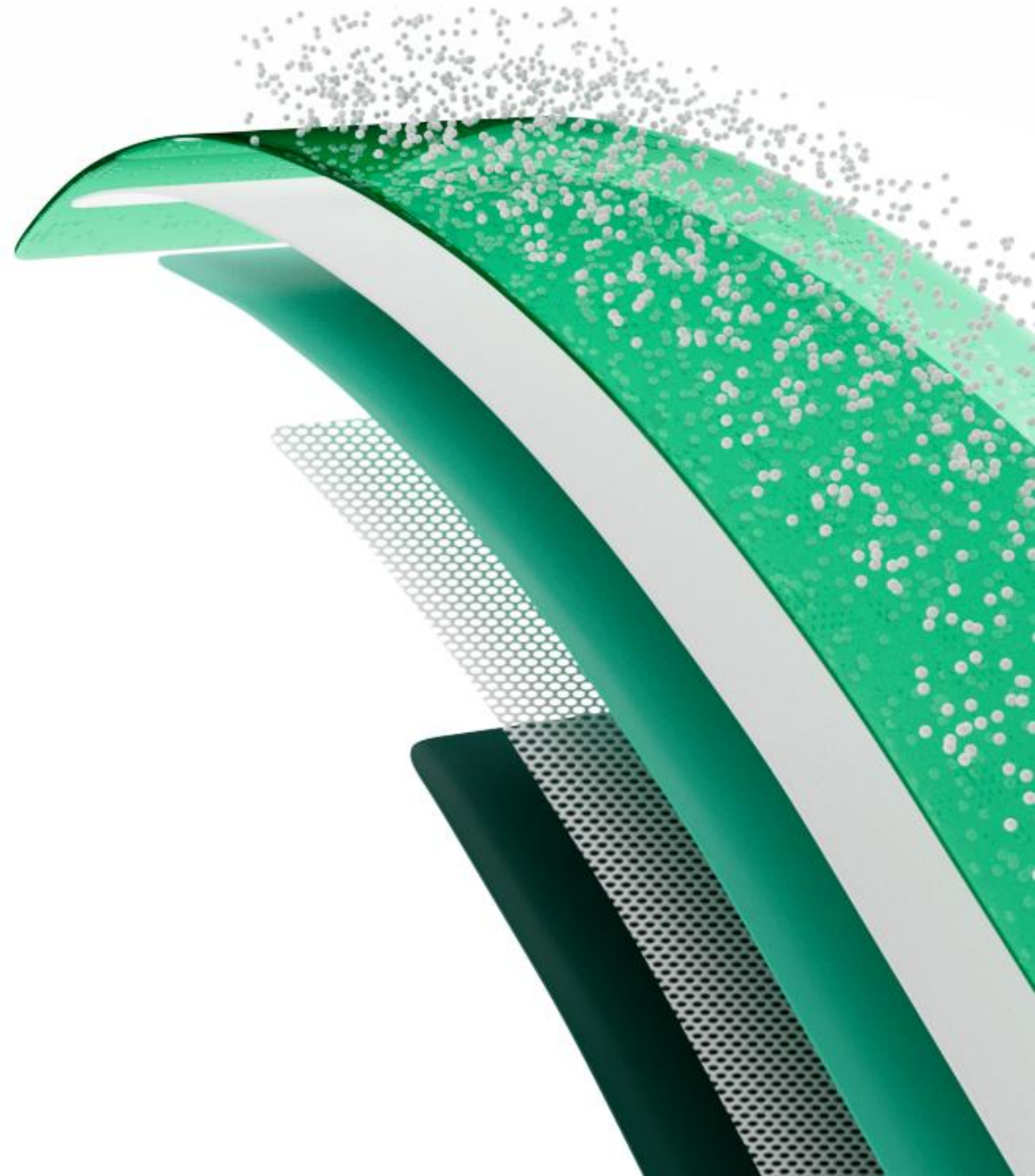


Microsoft Fabric

A unified analytics solution for the era of AI



Capacity Management





SESSION AGENDA

01

Introduction to Microsoft Fabric

02

Fabric Capacity

03

Smoothing & Bursting

04

Overloading and Throttling

05

Capacity Mitigations

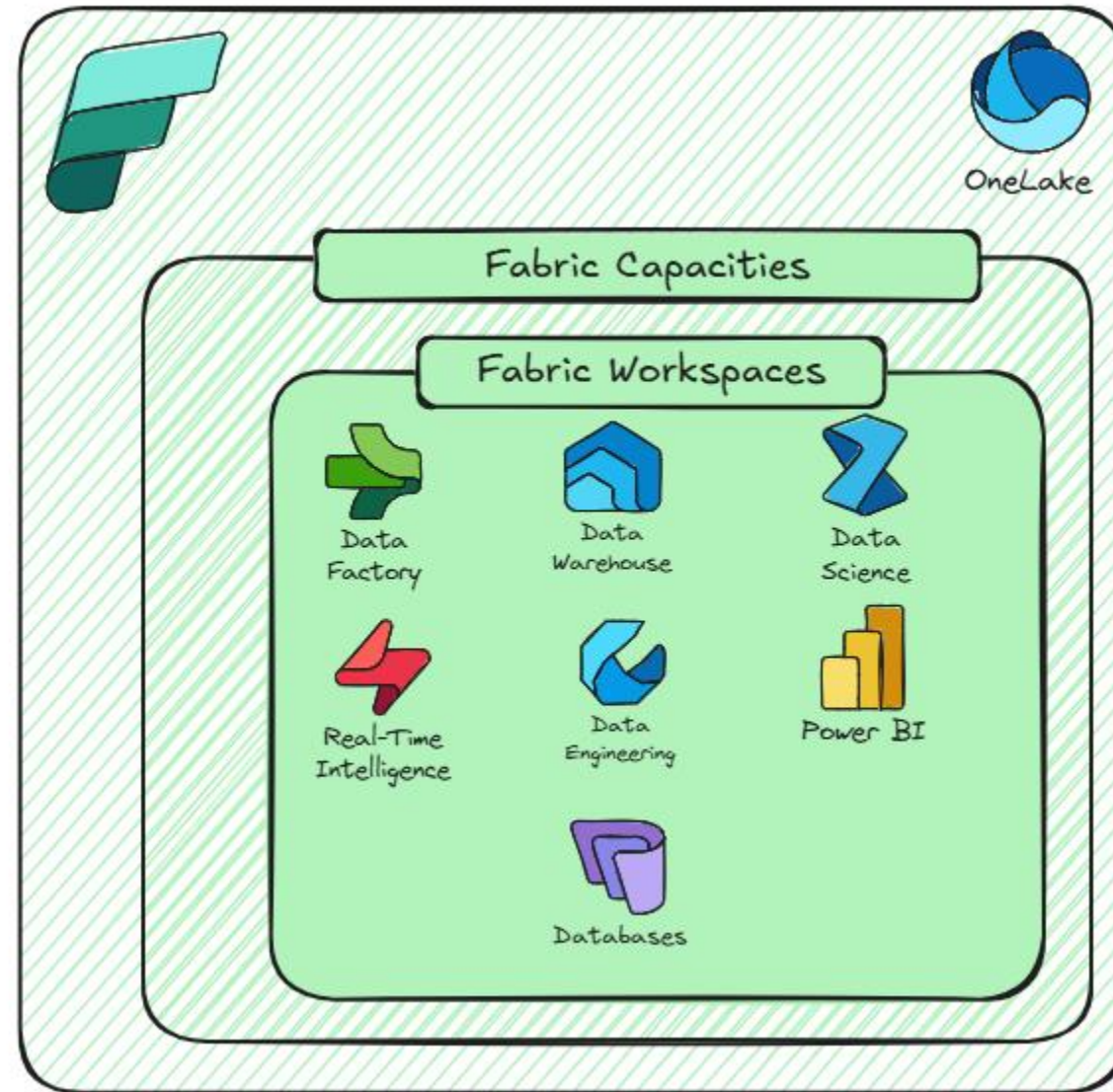
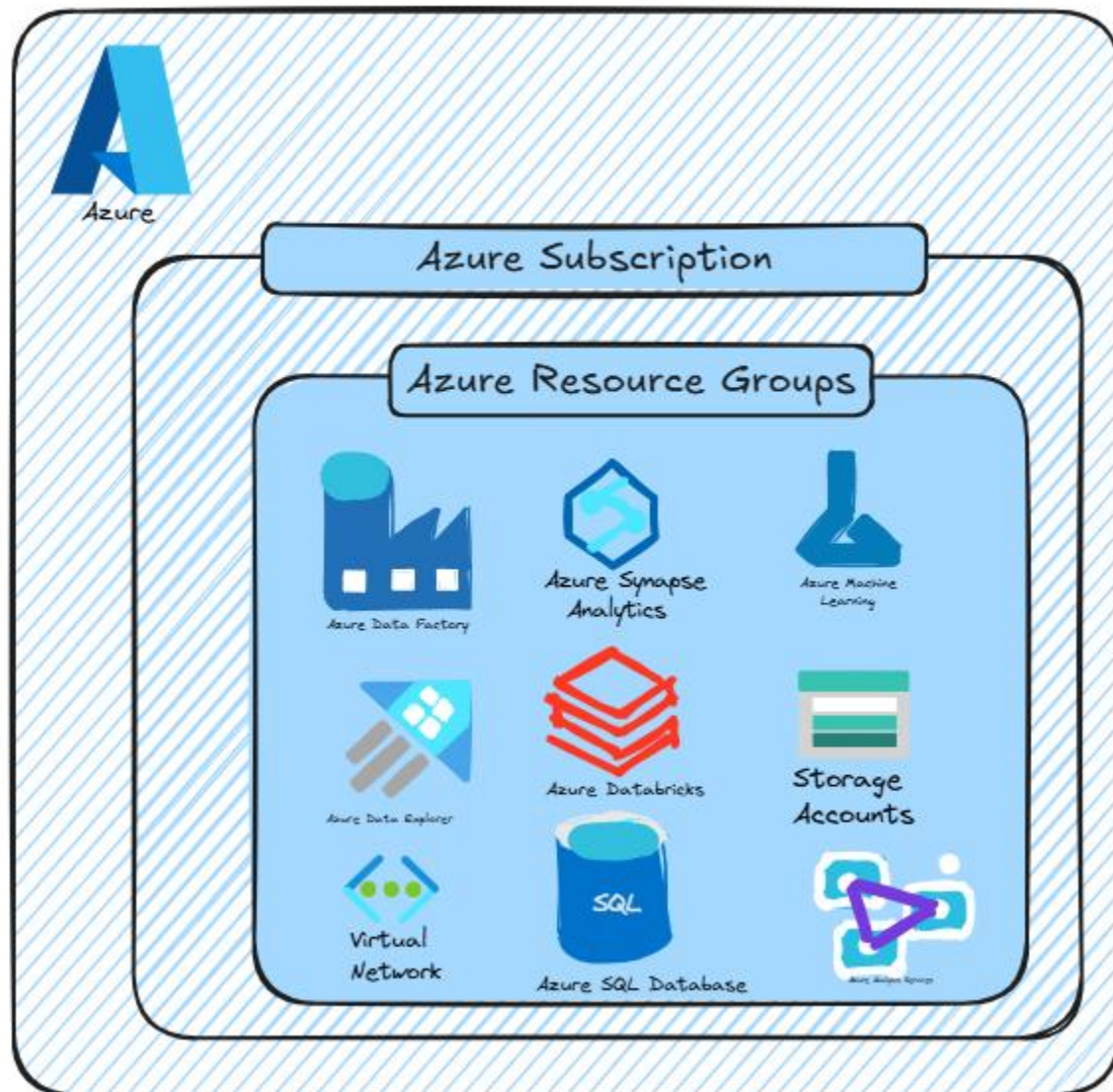
06

Demo - Tour of Microsoft Fabric & Microsoft Fabric Capacity

Introduction to Microsoft Fabric

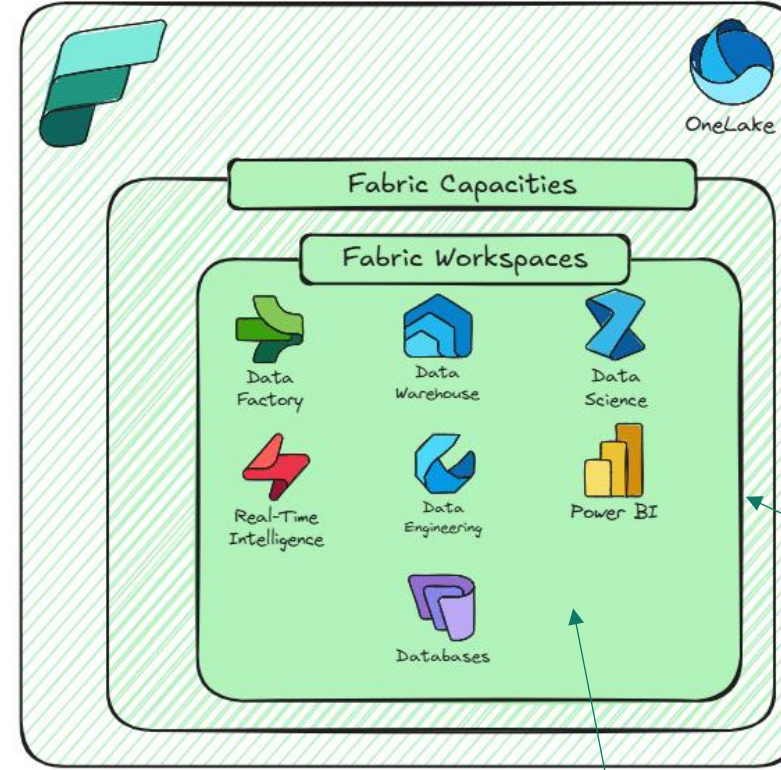
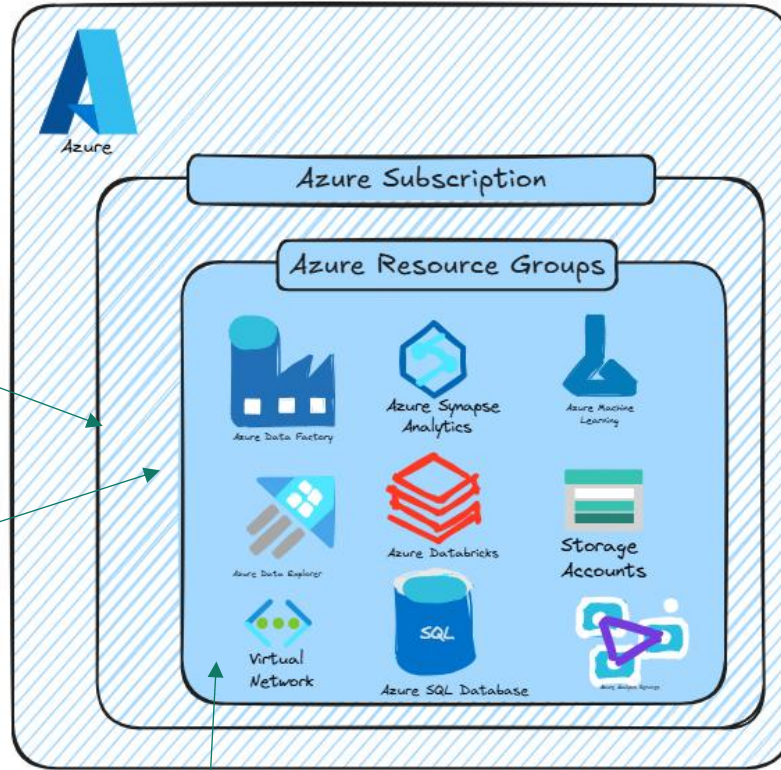


Similar Concepts, but different architectures



It's not PaaS

It's SaaS!



Billing Construct
Agnostic of RGs

Resource Container
Access Control
Agnostic of Resources

Compute Provisioned and Scaled Independently
Contains Infra and Network Services

Platform Services
Tenant Governance
Infra Components

Billing Construct
SKU Limit
Governance

Resource Container
Access Control
Workspace Settings

Workloads Scale as a Unit



Fabric foundation

- Microsoft Fabric is built on the SaaS foundation of the Power BI service:

- Authentication and authorization model
- Security and data protection controls
- Configuration and administration tools
- Content management experience
- Navigational experience
- Content discoverability and lineage
- Governance tools
- Auditing and monitoring tools

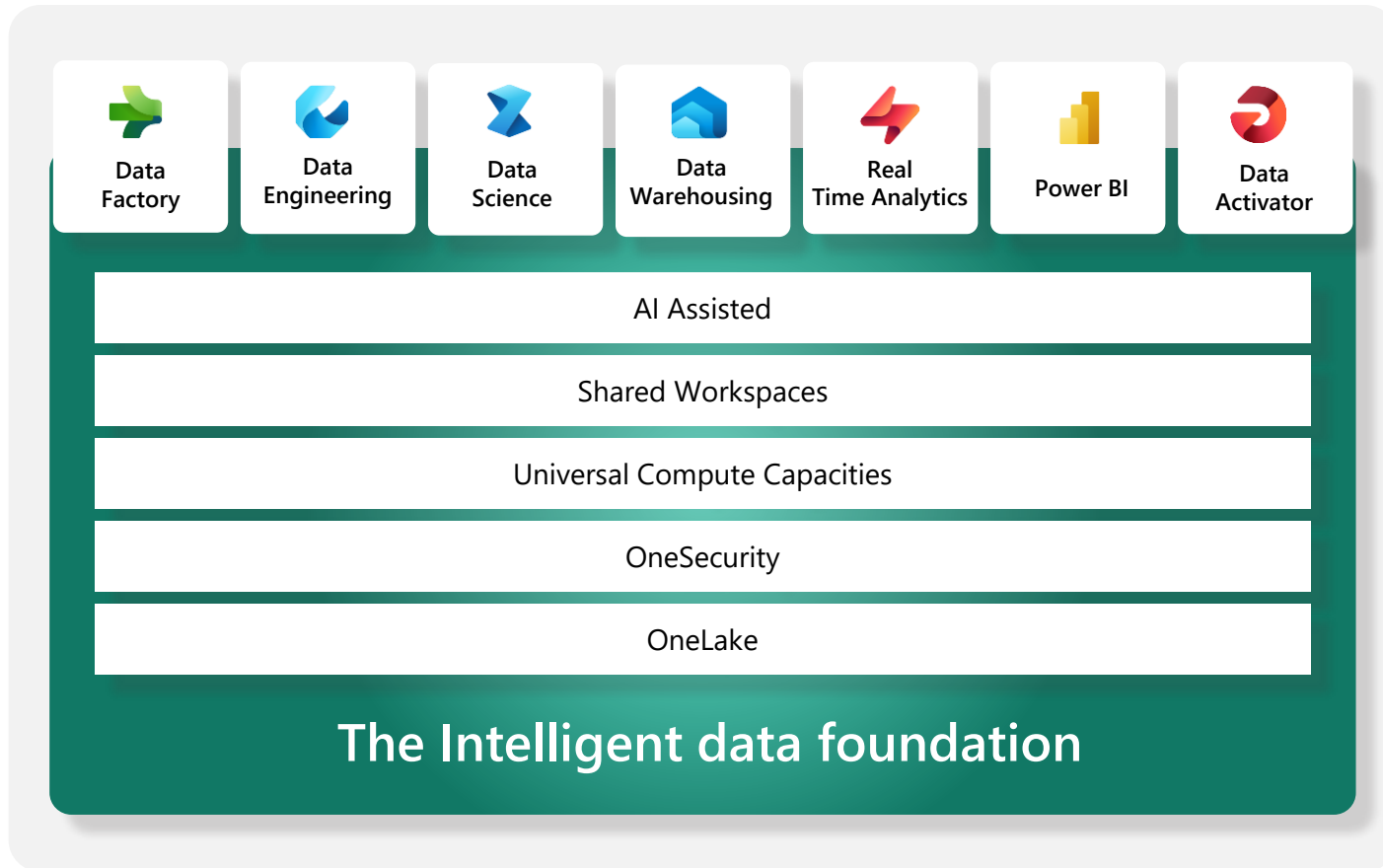


- Fabric governance is an extension of Power BI governance (with broader scope)
- Power BI admins are already familiar with core principles of Fabric governance
- Power BI admin role will become a Fabric Admin role



Microsoft Fabric - a unified solution

The data platform for the era of AI



Single...

Onboarding and trials
Sign-on
Navigation model
UX model
Workspace organization
Collaboration experience
Data Lake
Storage format
Data copy for all engines
Security model
CI/CD
Monitoring hub
Data Hub
Governance & compliance

An end-to-end analytics platform that brings together all the data and analytics tools that organizations need to go from the data lake to the business user



Data Engineers

- **Execute faster** with the ability to spin up a Spark VM cluster in seconds, or configure with familiar experiences like Git DevOps pipelines for data engineering artifacts
- **Streamline your work** with a single platform to build and operate real-time analytics pipelines, data lakes, lake houses, warehouses, marts, and cubes using your preferred IDE, plug-ins, and tools.
- **Reduce costly data replication** and movement with the ability to produce base datasets that can serve data analysts and data scientists without needing to build pipelines

Serve data via
warehouse or
lakehouse

Supporting experiences:



Data Factory



Data Warehouse



Data Engineering



Real-time analytics



Data Scientists

- **Quickly tune a custom model** by integrating a model built and trained in Azure ML in a Spark notebook
- **Work faster** with the ability to use your preferred data science frameworks, languages, and tools
- **Bypass engineering dependencies** with the ability to use your preferred no-code ML Ops to deploy and operate models in production
- **Tap into proven-at-scale models and services** to accelerate your AI differentiation (AOAI, Cognitive Services, ONNX integration, etc).

Serve
transformed
data

Supporting experiences



Data Science



Azure ML



Data Analysts

- **Avoid slow, progress-stagnating data wrangling** by seamlessly triggering a workflow that can unlock data engineering tools and capabilities quickly.
- **Accelerate your work** with visual and SQL based tools for self-serve data transformations and modeling as well as self-serve tools for reporting, dashboards, and data visualizations
- **Turn data into impact** with industry-leading BI tools and integration with the apps your people use everyday like Microsoft 365

Serve insights
via
embedding

Supporting experiences



Data Warehouse



Real-time analytics



Power BI



Data Citizens

- **Make more data-driven decisions** with actionable insights and intelligence in your preferred applications
- **Maintain access to all the data you need**, without being overwhelmed by data ancillary to your role thanks to fine grain data access management controls

Supporting experiences



Power BI



Microsoft 365

Serve data via warehouse or lakehouse



Data Stewards

- **Maintain visibility and control of costs** with a unified consumption and cost model that provides evergreen spend optics on your end-to-end data estate
- **Gain full visibility and governance** over your entire analytics estate from data sources and connections to your data lake, to users and their insights

Governance security & reliability in Fabric – vision



Microsoft Fabric capabilities

Microsoft Fabric governance and compliance encompasses a range of capabilities designed to enable organizations to understand, protect, manage, and monitor sensitive information.

Data Protection

- Sensitivity labels
- Data loss prevention
- Endorsement

Data Management

- Metadata scanning
- Data lineage and impact analysis
- Domains

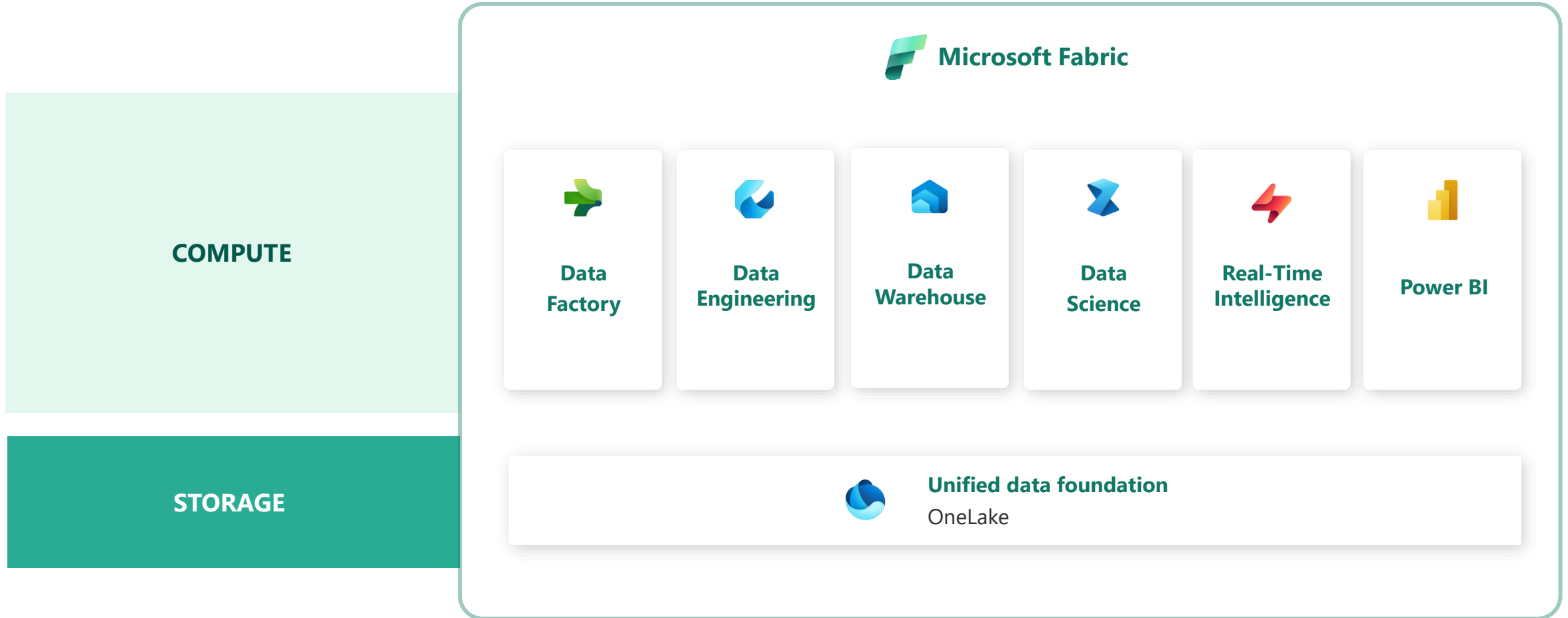
Monitoring & Compliance

- Purview hub integration
- Certifications
- Audit

Fabric capacity



Microsoft Fabric business model overview



Microsoft Fabric simplicity

Rather than provisioning and managing separate compute for each workload, with Microsoft Fabric, your bill is determined by two variables: the amount of compute you provision and the amount of storage you use.

COMPUTE

A shared pool of capacity that powers all capabilities in Microsoft Fabric, from data modeling and data warehousing to business intelligence.

Pay-as-you-go (per sec billing with one minute minimum).

STORAGE

A single place to store all data.
Pay-as-you-go (\$ per GB / month).

How it works

Capacity units (CUs) = Compute power

Capacity units (CUs) are units of measure that represent a pool of compute power needed. Compute power is required to run queries, jobs, or tasks.

CU Consumption

CUs consumed by data warehousing include read and write activity against the Warehouse, and read activity on the SQL Endpoint of the Lakehouse

1 Fabric capacity unit = 0.5 Warehouse vCores

For example, a Fabric capacity SKU F64 has 64 capacity units, which is equivalent to 32 Warehouse vCores

Sample of consumption rates:

F64 SKU provides a set of 64 capacity units

Capability	CUs utilized for each processing time
Data Warehouse	2
Power BI	8
Spark	0.5
Dataflows	16

This means a Data Warehousing job will utilize 2 CUs while a Power BI query will utilize 8 CUs at the same processing time.

Fabric capacity pricing

Fabric capacity is priced uniquely across regions. The following table shows the pricing at US West 2 for your reference. Fabric capacity can be purchased at Azure portal.

1 CU pay-as-you-go price

\$0.18 / hour

SKU	Capacity unit (CU)	Pay-as-you-go (hourly) at US West 2	Pay-as-you-go (monthly) at US West 2	Reservation at US West 2 (~41% savings)
F 2	2	\$0.36	\$262.8	\$156.334
F 4	4	\$0.72	\$525.6	\$312.667
F 8	8	\$1.44	\$1,051.2	\$625.334
F 16	16	\$2.88	\$2,102.4	\$1,250.667
F 32	32	\$5.76	\$4,204.8	\$2,501.334
F 64	64	\$11.52	\$8,409.6	\$5,002.667
F 128	128	\$23.04	\$16,819.2	\$10,005.334
F 256	256	\$46.08	\$33,638.4	\$20,010.667
F 512	612	\$92.16	\$67,276.8	\$40,021.334
F 1024	1024	\$184.32	\$134,553.6	\$80,042.667
F 2048	2048	\$368.64	\$269,107.2	\$160,085.334

Note:

1. Monthly rate calculation: F2 $\$0.18 \times 2 \times 730 = \262.8

2. Power BI Pro license requirement: (an existing requirement in Power BI Premium per capacity)

- Consumption of Power BI reports requires Pro license for F2-F32
- Publishing/authoring requires Pro license for all SKUs

OneLake storage pricing

OneLake is a data lake built into Microsoft Fabric and provides a single place to store all organizational data. Data storage is charged at a rate of \$ per GB per month and priced uniquely across regions.



Data Storage

Type	Pay-as-you-go price at France Central
OneLake storage	0.0216 € per GB/month
OneLake BCDR storage	0.0389 € per GB/month
OneLake cache	0.2247 € per GB/month

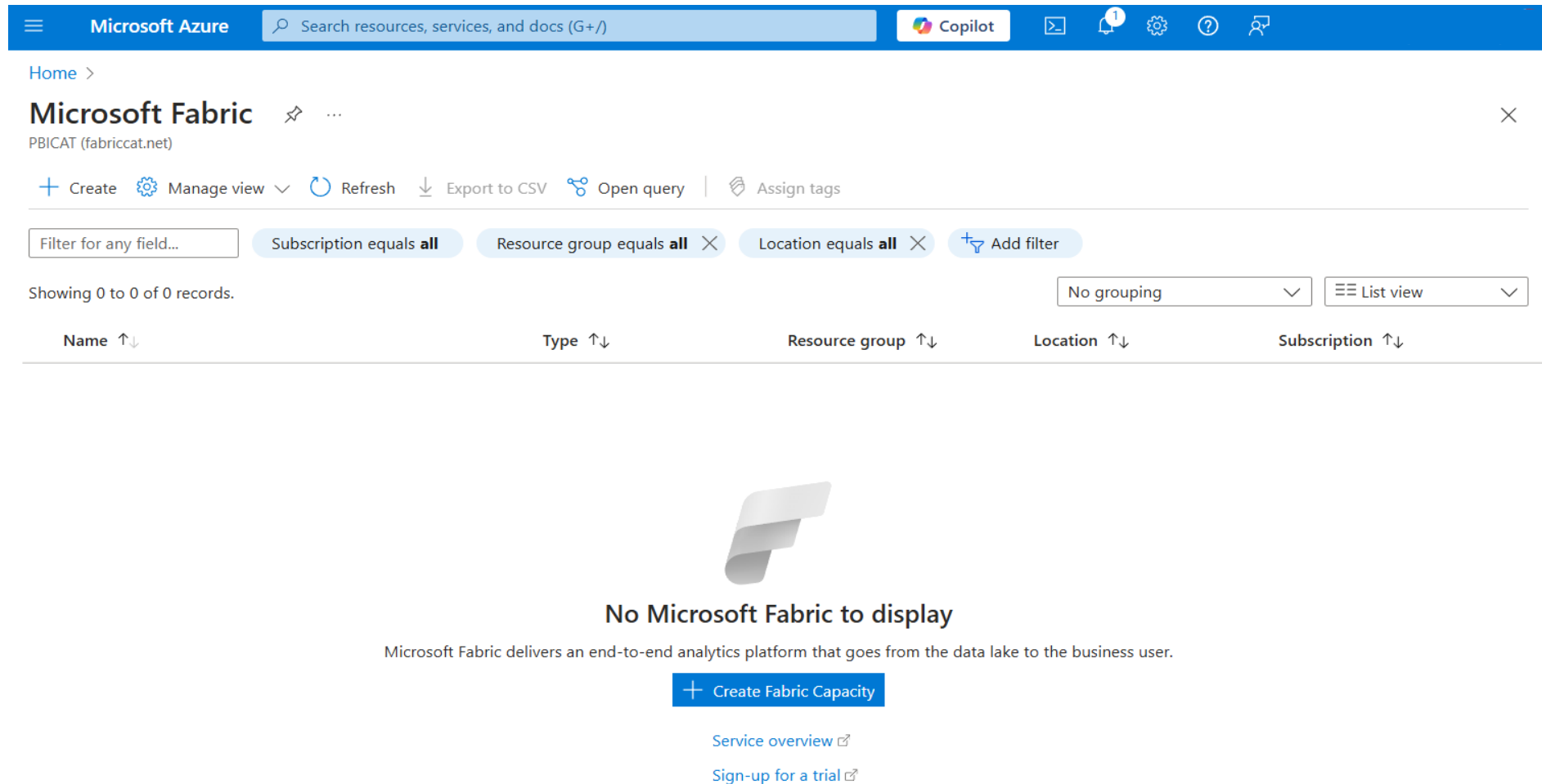


Data Transfer & Internet Egress

Cross-region data transfer network charges may apply based on source/destination of each storage access. Learn more at the [Bandwidth Pricing](#).

Purchasing capacity

Azure Portal, need permission to purchase Fabric



The screenshot shows the Microsoft Azure portal interface. At the top, the navigation bar includes the Microsoft Azure logo, a search bar, and various utility icons. Below the navigation bar, the breadcrumb trail shows 'Home >'. The main heading is 'Microsoft Fabric', with a subtext 'PBICAT (fabriccat.net)'. Below this, there are action buttons: '+ Create', 'Manage view', 'Refresh', 'Export to CSV', 'Open query', and 'Assign tags'. A filter bar shows 'Subscription equals all', 'Resource group equals all', and 'Location equals all', with an 'Add filter' button. Below the filter bar, it says 'Showing 0 to 0 of 0 records.' and has dropdowns for 'No grouping' and 'List view'. A table header is visible with columns: 'Name', 'Type', 'Resource group', 'Location', and 'Subscription'. Below the table, there is a large graphic of the Microsoft Fabric logo and the text 'No Microsoft Fabric to display'. Below this, a description states: 'Microsoft Fabric delivers an end-to-end analytics platform that goes from the data lake to the business user.' At the bottom, there is a blue button labeled '+ Create Fabric Capacity' and two links: 'Service overview' and 'Sign-up for a trial'.

Home >

Microsoft Fabric

PBICAT (fabriccat.net)

+ Create Manage view Refresh Export to CSV Open query Assign tags

Filter for any field... Subscription equals all Resource group equals all Location equals all Add filter

Showing 0 to 0 of 0 records. No grouping List view

Name	Type	Resource group	Location	Subscription
------	------	----------------	----------	--------------



No Microsoft Fabric to display

Microsoft Fabric delivers an end-to-end analytics platform that goes from the data lake to the business user.

+ Create Fabric Capacity

[Service overview](#)

[Sign-up for a trial](#)

Microsoft Fabric Trial

Pre-requisites & Power BI Account

Option 1- **Microsoft 365 Developer Account** -

With Admin right: [Click Here to Create One](#) if you do not have one. This license give you access to Power BI Pro account.

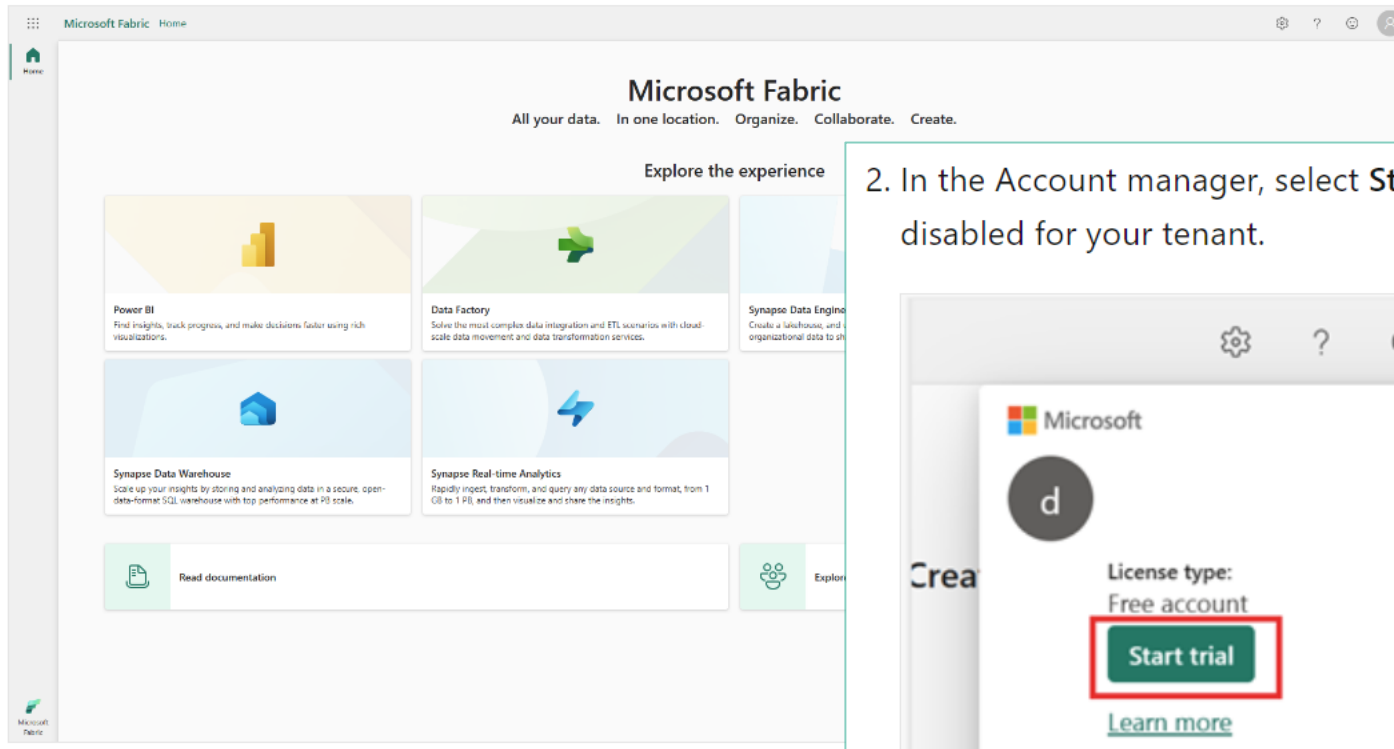
Option 2- **Microsoft Fabric Free Trial:** You need Microsoft Fabric Free trial to gain access to functionalities in Microsoft Fabrics. Click on the link below to get one <https://app.fabric.microsoft.com>

Option 3- If you're an existing Power BI user, you can skip to [Start the Fabric trial](#)

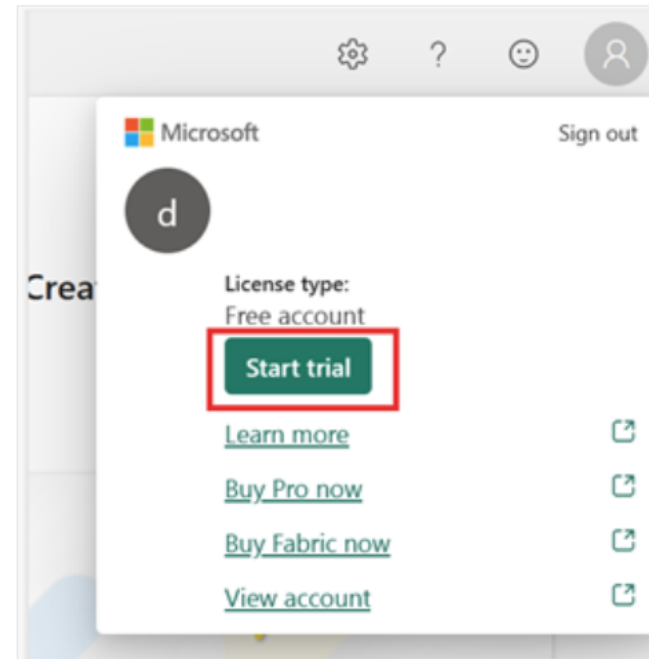


Start the Fabric trial

1. Open the [Fabric homepage](#) and select the Account manager.



2. In the Account manager, select **Start trial**. If you don't see the *Start trial* button, trials might be disabled for your tenant.



Capacities are a shared resource

Shared across workloads

A single capacity is providing the compute power for all the Fabric's workloads.

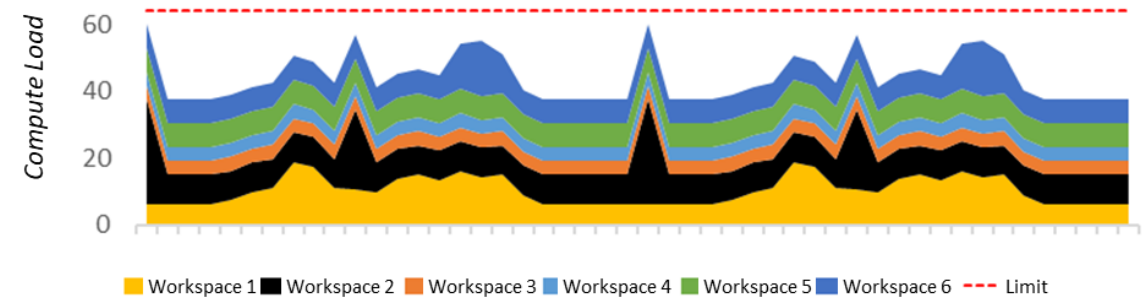
There is no need to allocate compute for each workload separately.



Shared Across Projects

A single capacity typically supports dozens of separate projects simultaneously, each managed in its own workspace.

It is rare to have a capacity dedicated to a single project, although that is possible when needed



Shared across users

For each project, many developers will share a workspace where collaborative development and consumption at scale is managed.

Each creator can provision any artifact and run any job without the need for any pre-approval or planning

Developers/Creators

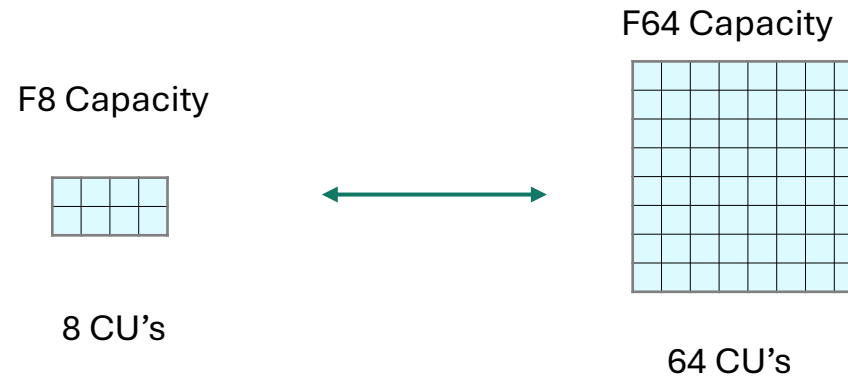


Consumers

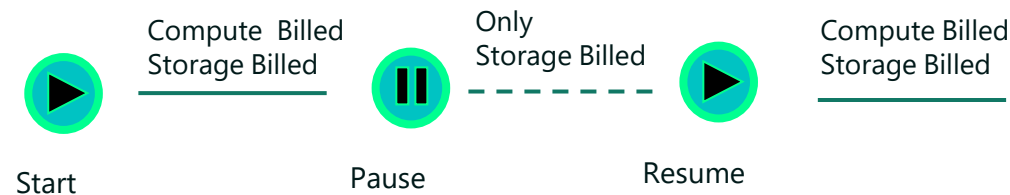


Fabric Capacities are flexible and scalable

Resize to increase or decrease the SKU to meet your compute needs.



Pause and Resume the capacity



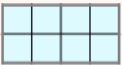
Multiple capacities for one tenant

Capacities can be configured to meet scale, usage and governance requirements while tuning to minimize TCO and performance goals

Scale Vertically

Increased capacity size provides more throughput

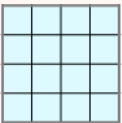
F8 Capacity



8 CU's



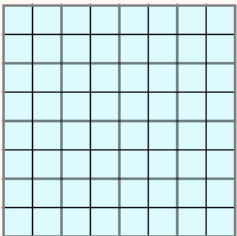
F16 Capacity



16 CU's



F64 Capacity

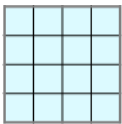


64 CU's

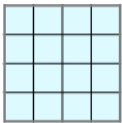
Scale Horizontally

Scale horizontally using the benefits of modular design for hardened isolation and governance

Development

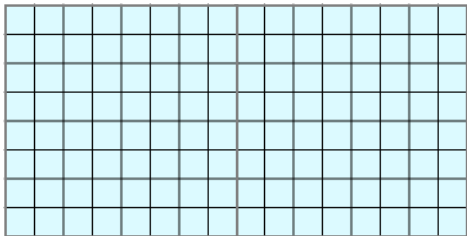


Test / "Tryout"



F16 Capacities

Prod

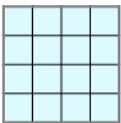


F128 Capacity

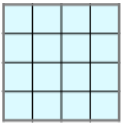
Regional Availability

Use different capacities for different regions to support GDPR / Data residency requirements

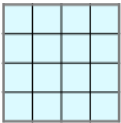
Region: UK South



Region: France Central



Region: Sweden Central



F16 Capacities

Smoothing & Bursting



Great performance and simplified management

Fabric uses intelligent techniques on all capacities:



Smoothing



Bursting

What is Smoothing?

Load stabilization

Load smoothing helps capacities self-stabilize by flattening large spikey loads into a smooth load profile, eliminating temporal spikes

Scheduling contention elimination

Large/scheduled Jobs are smoothed over 24 hours, eliminating the need to decide the timing and order of job execution

Operations are always executed for maximum perf

Bad actor protection

Interactive operations smoothed over several minutes, preventing a single user with a very demanding query from hogging the entire capacity

What is Bursting?

Job acceleration

Bursting provides extra compute resources beyond what have been purchased, to jobs and queries to accelerate their completion

Go beyond

The extra resources of bursting allow jobs to **utilize far more resources than “face value”**

Instead of running a job on 64 CU and completing in 60 seconds, bursting could use 256 CUs to complete the job in 15 seconds

Same amount of work, just completed faster

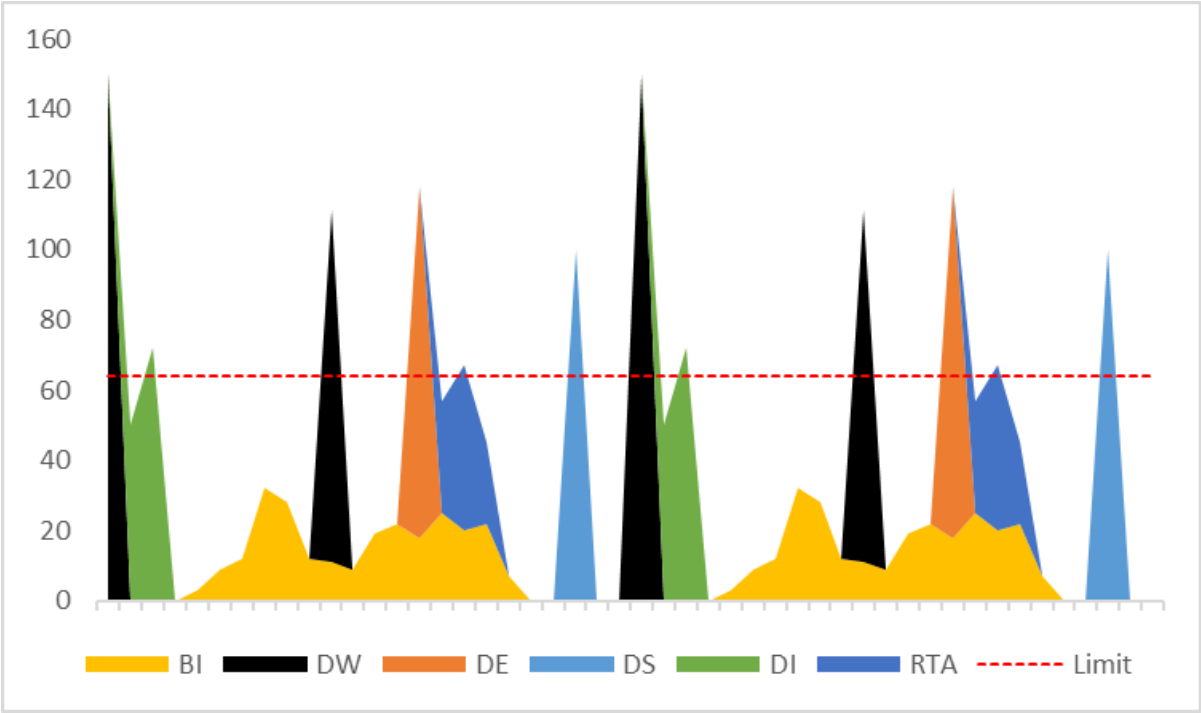
No hassle, No overload

Bursting is automatic when the system reasons it can accelerate the job by applying extra resources. No settings are required

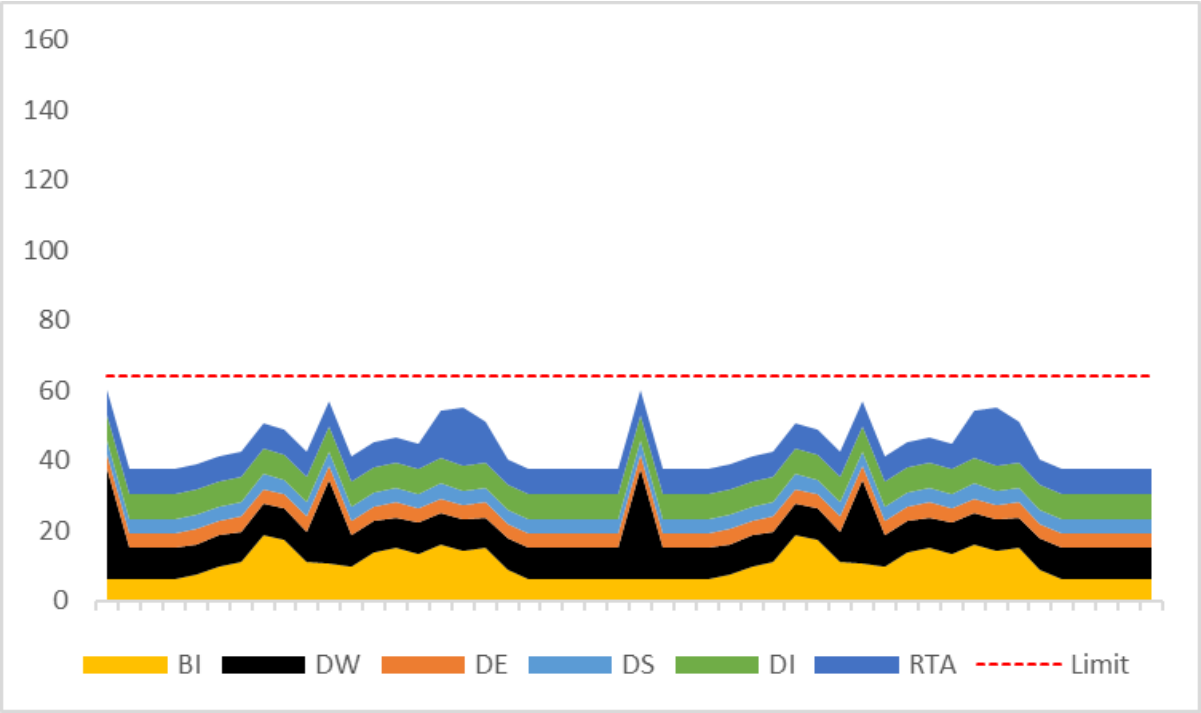
Bursting never causes an overload as the *smoothing* mechanism will always flatten the resource burst

Bursting and Smoothing

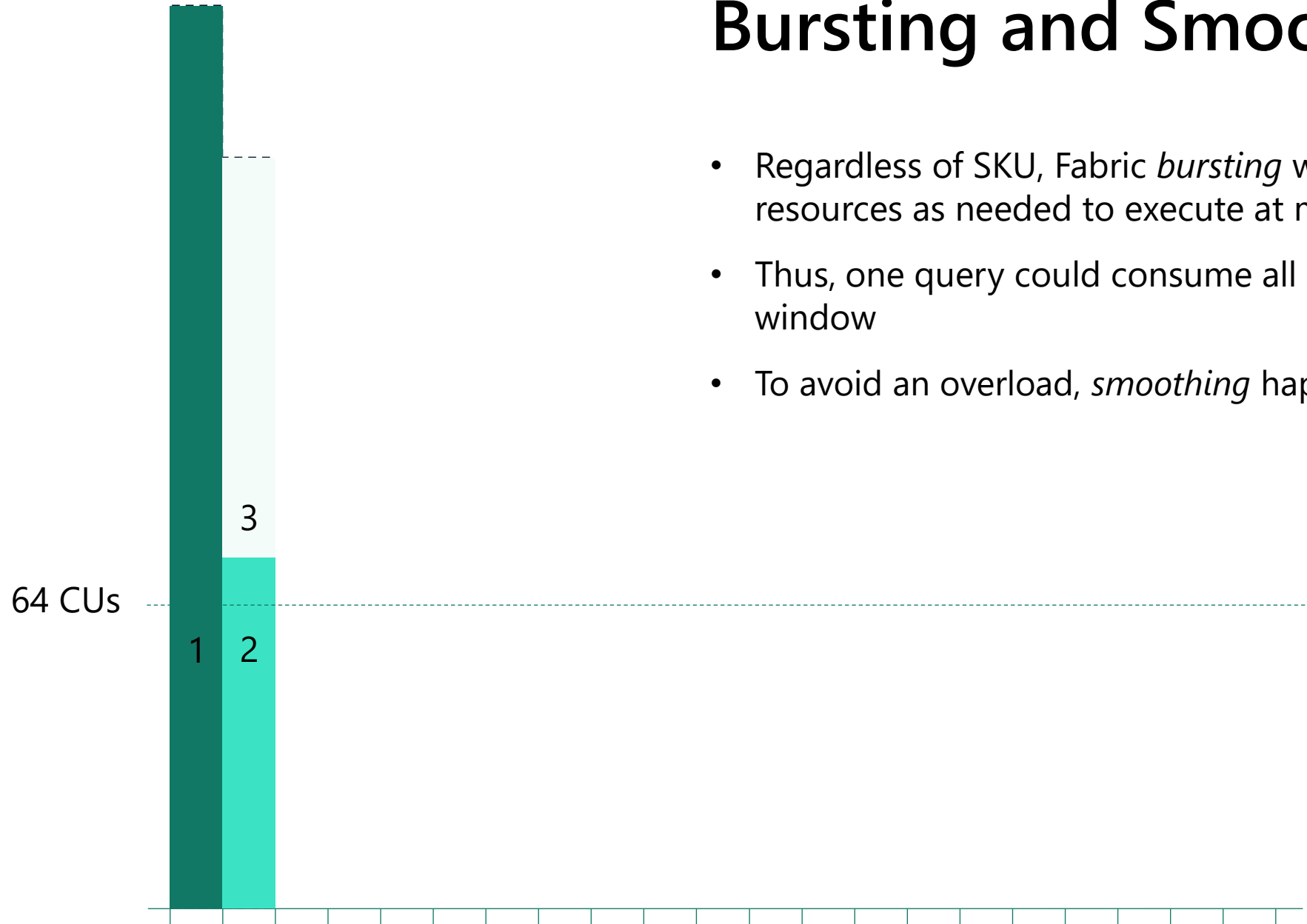
Before Smoothing



After Smoothing

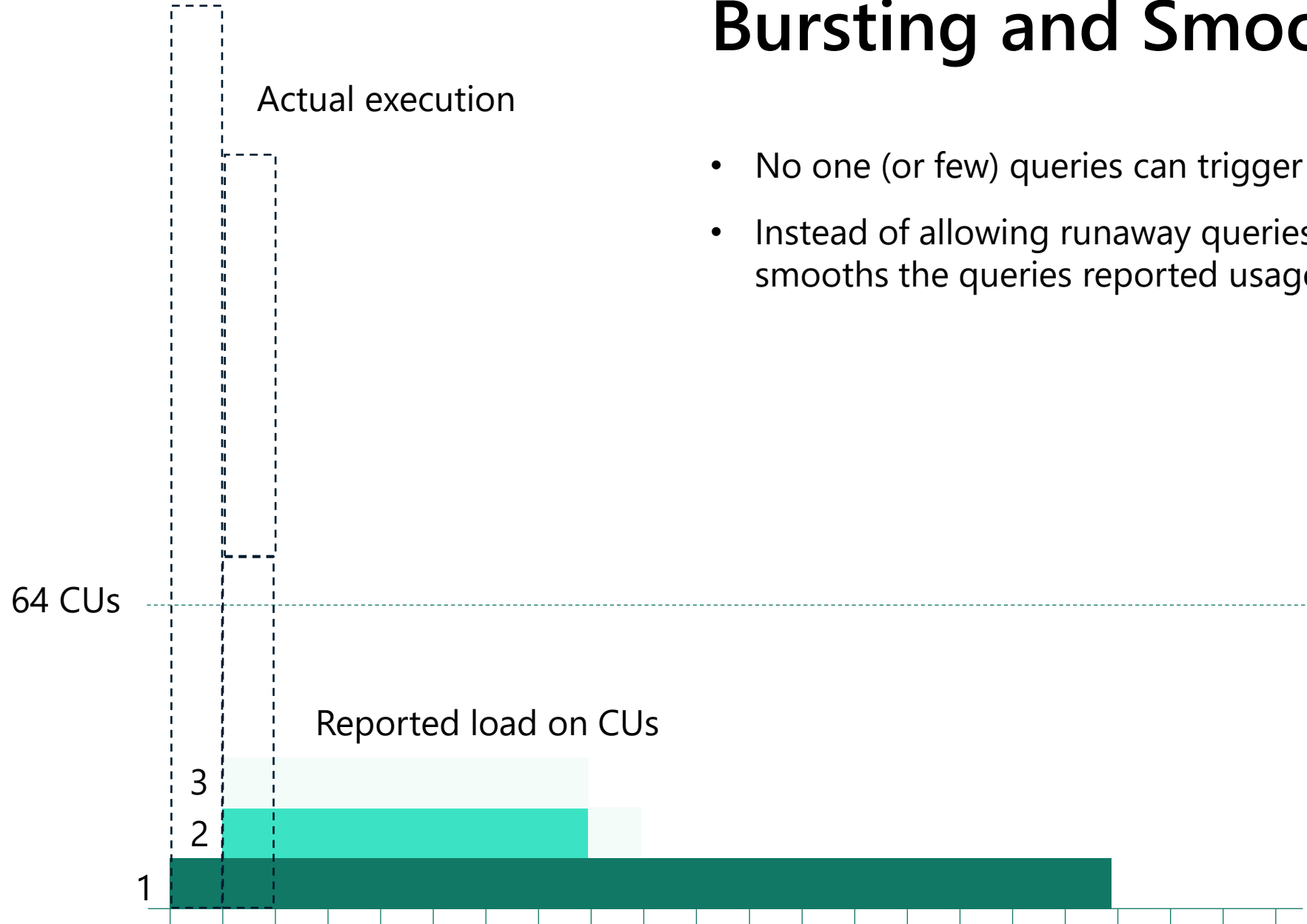


Bursting and Smoothing



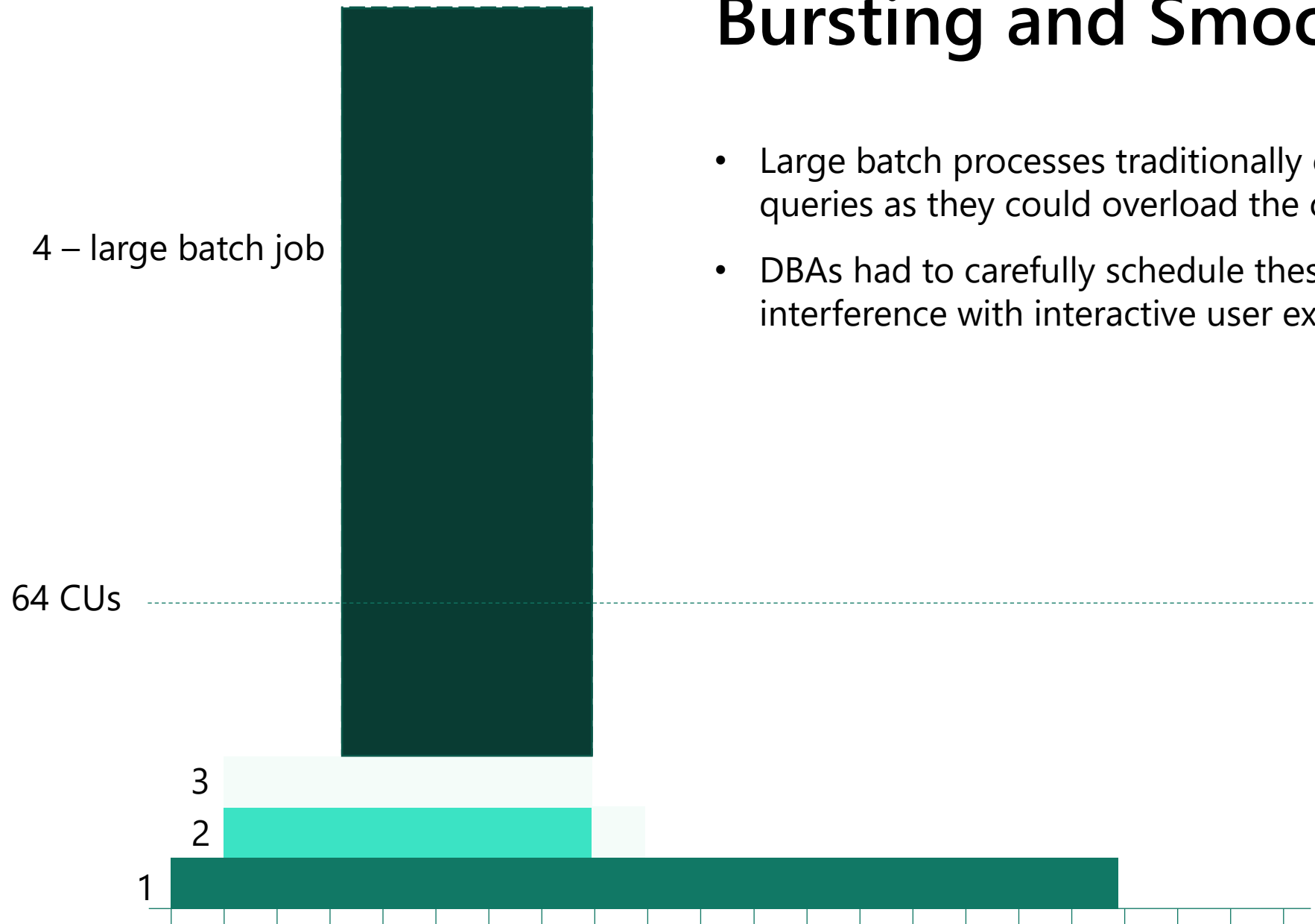
- Regardless of SKU, Fabric *bursting* will automatically allocate resources as needed to execute at maximum performance
- Thus, one query could consume all the quota of a single time window
- To avoid an overload, *smoothing* happens

Bursting and Smoothing



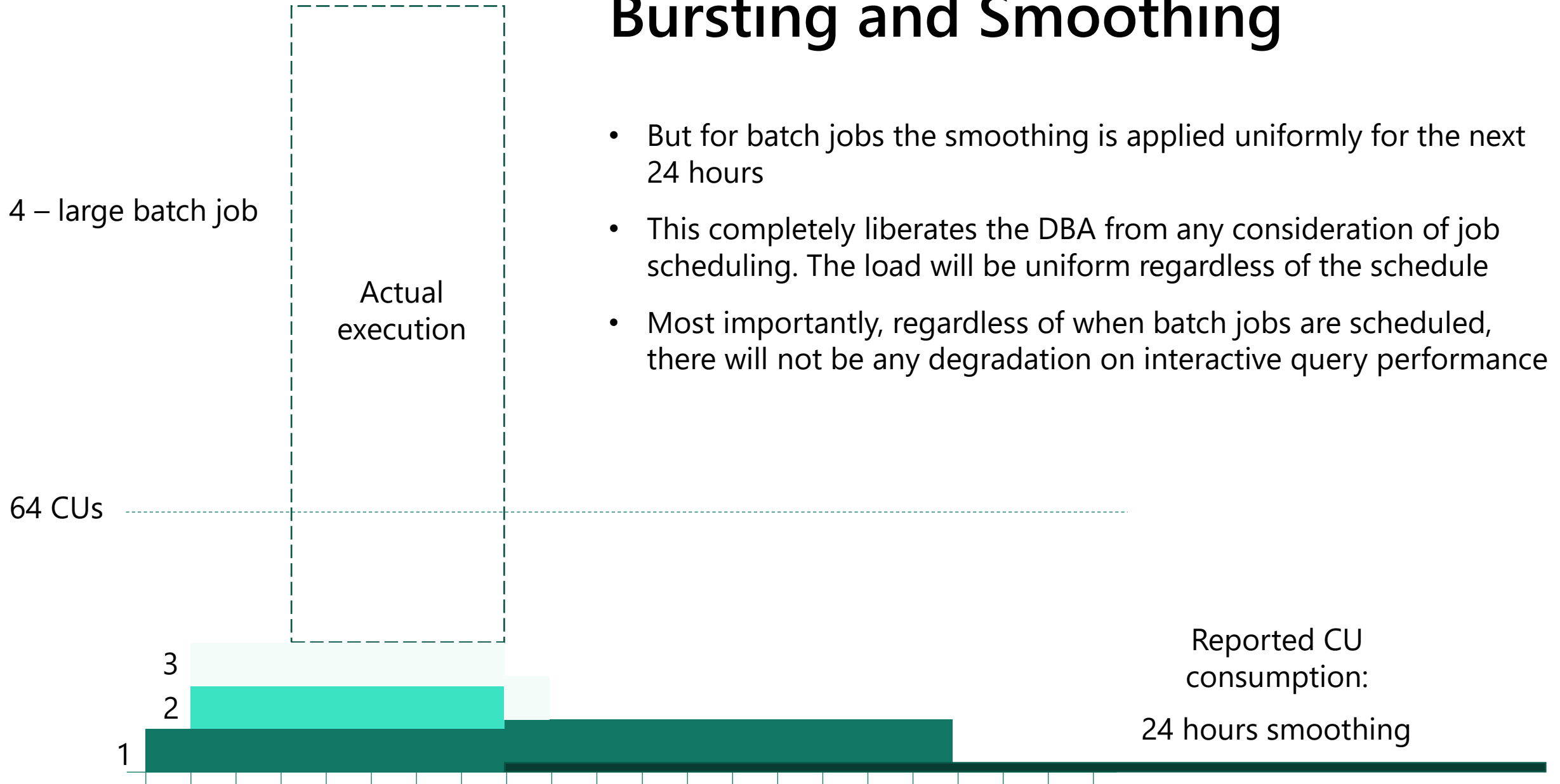
- No one (or few) queries can trigger an overload
- Instead of allowing runaway queries to create a overload, Fabric smooths the queries reported usage to future time windows

Bursting and Smoothing



- Large batch processes traditionally compete with interactive queries as they could overload the compute resource
- DBAs had to carefully schedule these jobs to off-hours to avoid interference with interactive user experiences

Bursting and Smoothing



- But for batch jobs the smoothing is applied uniformly for the next 24 hours
- This completely liberates the DBA from any consideration of job scheduling. The load will be uniform regardless of the schedule
- Most importantly, regardless of when batch jobs are scheduled, there will not be any degradation on interactive query performance



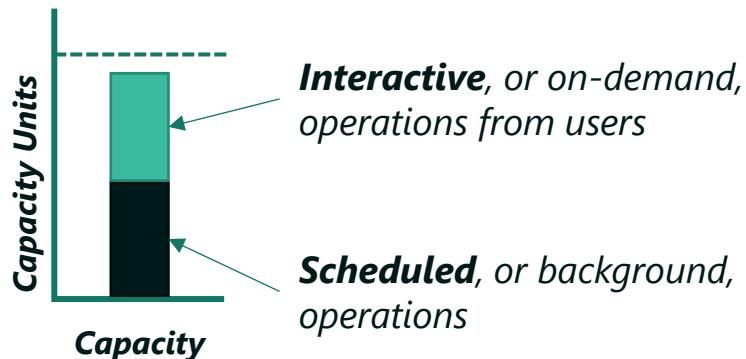
Overloading and throttling

What happens when there isn't enough?

Overloading happens when a sustained load from multiple jobs and user queries exceed the ability of the capacity to provide the compute required

Approaching the limit

As a capacity approaches its limit, warning messages and emails are sent to the capacity administrator

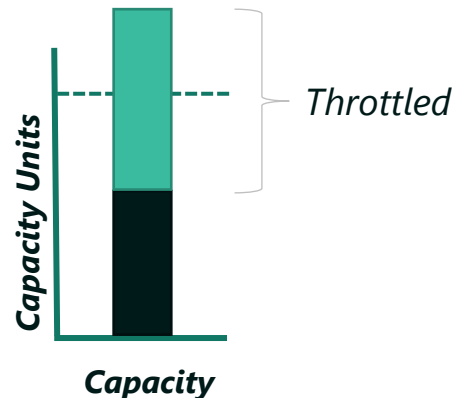


Exceeding the limit (overloaded)

When overloaded, two things happen:

Interactive operations from users still run, but with significant throttling delays degrading experiences

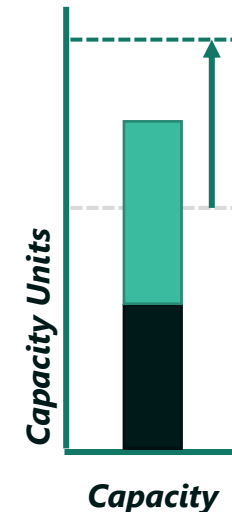
Scheduled operations continue to run at peak performance unless the overload is extreme



Remedy an overload

To remedy an overload, admins can:

- **Upgrade** the capacity to raise limits
- **Wait** until the overload state is over
- **Work with high load project owners** to reduce consumption
- Configure **proactive alerts**

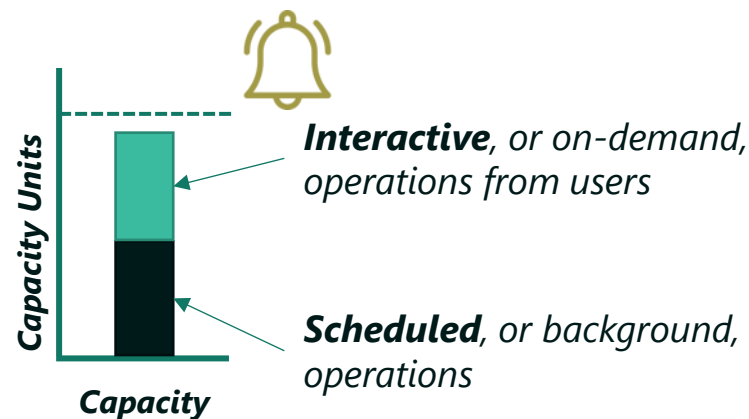


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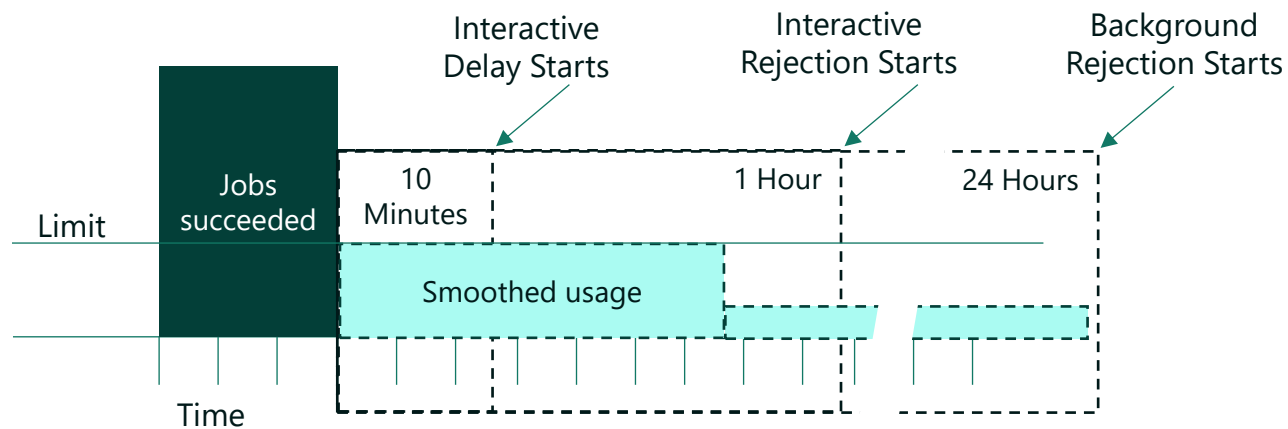
Exceeding the limit (overloaded)

When overloaded, two things happen:

- **Interactive requests** first are *delayed* and then are *rejected*
- **Background** requests are *rejected* if overload is extreme

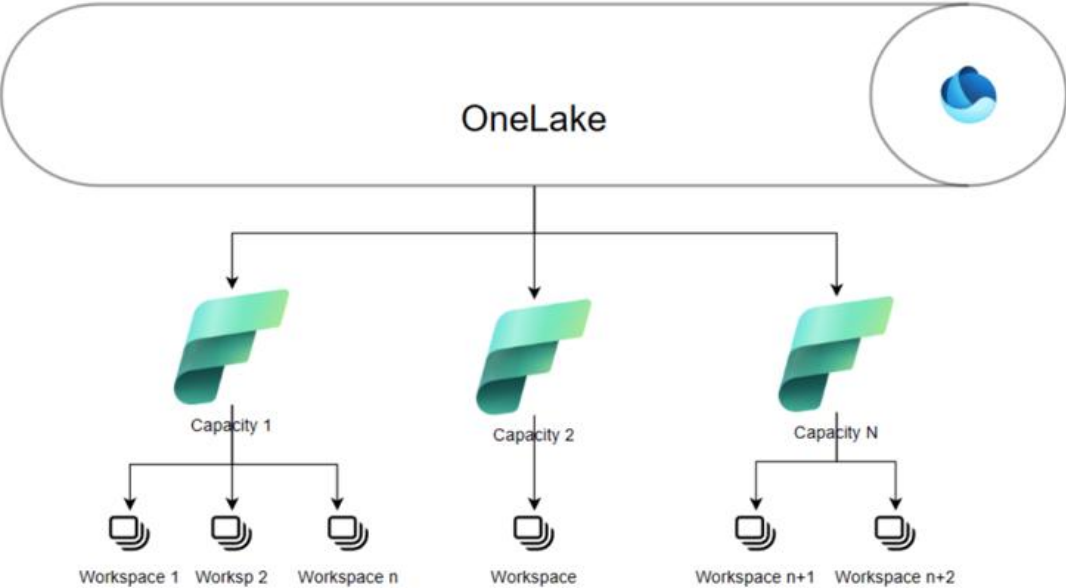
When throttling happens, users will see the **CapacityLimitExceeded** error.

Be aware that users may experience delays, slowness, or failures due to workload limits. These are unrelated to Capacity Throttling.



Capacity throttling policies

4 new capacity policies are outlined below.



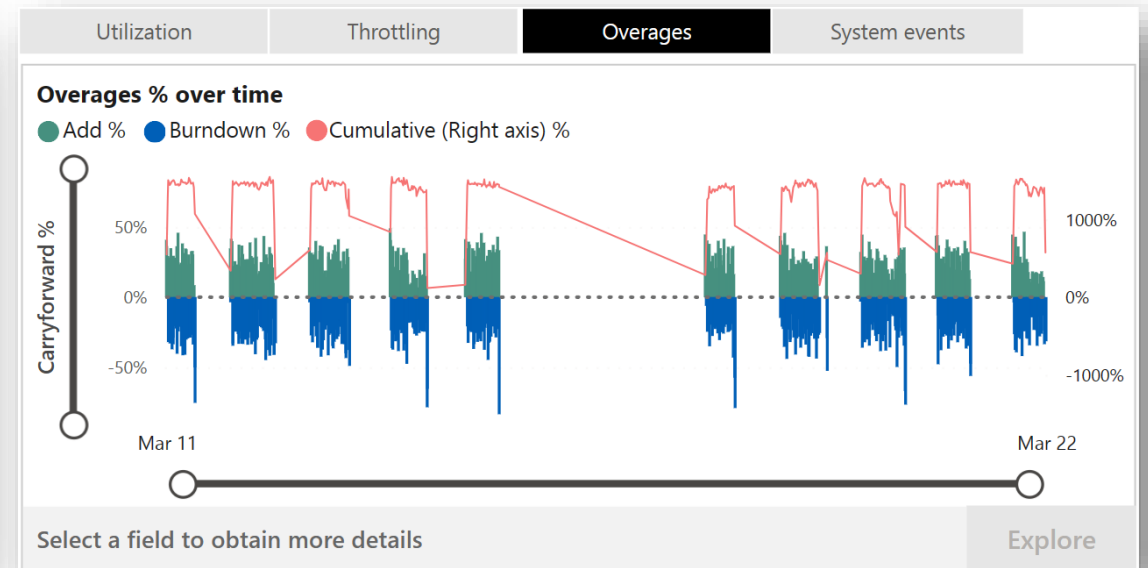
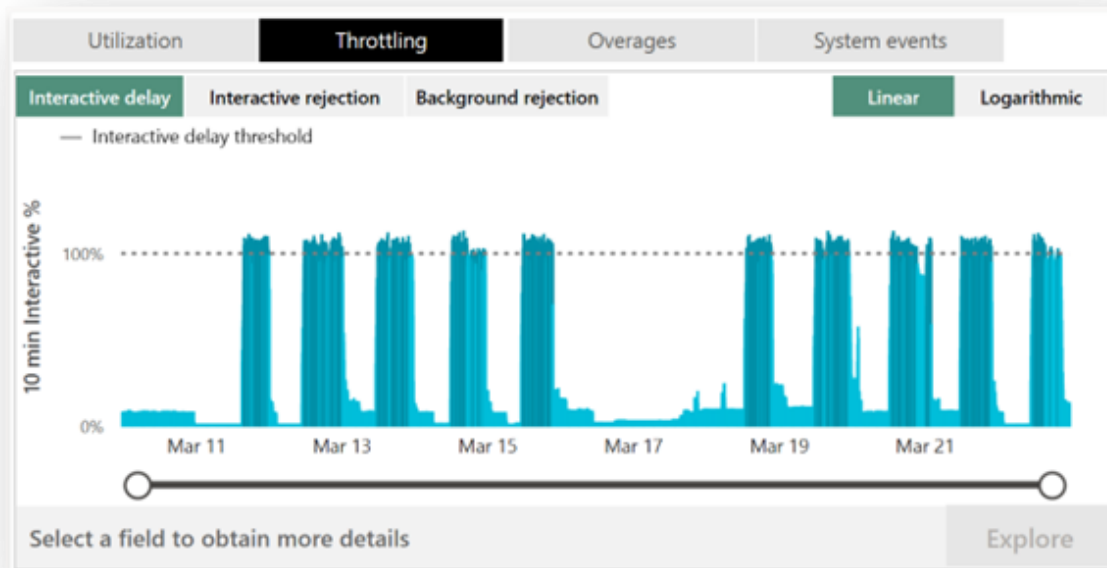
Usage	Policy Limits	Platform Policy Experience Impact
Usage <= 10 minutes	Overage protection	Jobs can consume 10 minutes of future capacity use without throttling.
10 minutes < Usage <= 60 minutes	Interactive Delay	User-requested interactive jobs are delayed 20 seconds at submission.
60 minutes < Usage <= 24 hours	Interactive Rejection	User-requested interactive jobs are rejected.
Usage > 24 hours	Background Rejection	All requests are rejected.

What happens when there isn't enough ?

Overloading happens when a sustained load from multiple jobs and user queries exceed the ability of the capacity to provide the compute required

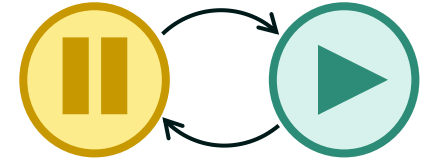
Exceeding the limit (overloaded)

When throttling is happening, you'll see it in the Capacity Metrics App.



How do you remedy an overload ?

Pausing and Resuming the Capacity



- 1) It helps manage compute costs.
- 2) It clears any overages** that have accumulated.
Use it to quickly resolve throttling.

Workloads stop
execution within
10min of Pause

New requests are
not allowed to start

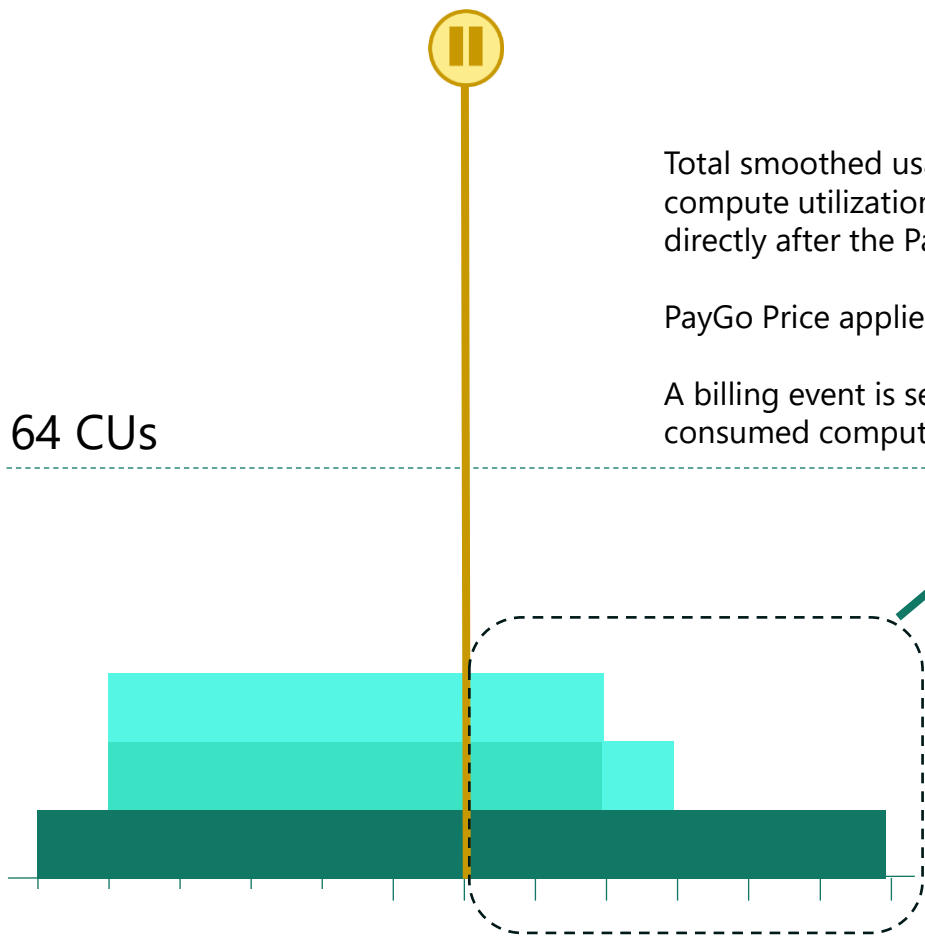
Smoothed usage will
be reconciled

Note: *OneLake storage costs continue to be billed while a capacity is paused*

How Capacity Pause & Resume works

When a capacity is **paused**...

Pause event on
Capacity

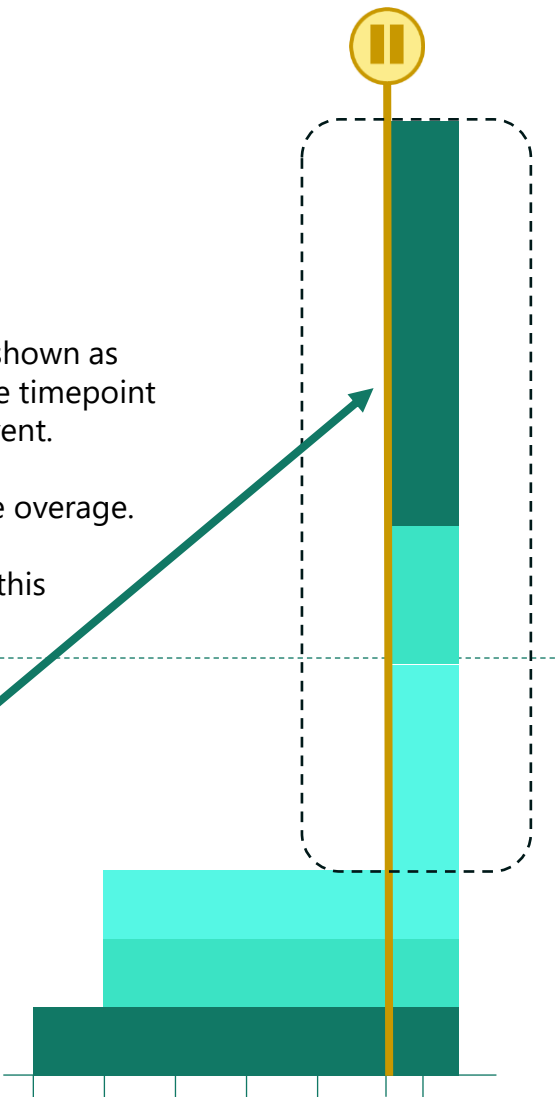


Total smoothed usage is shown as
compute utilization on the timepoint
directly after the Pause event.

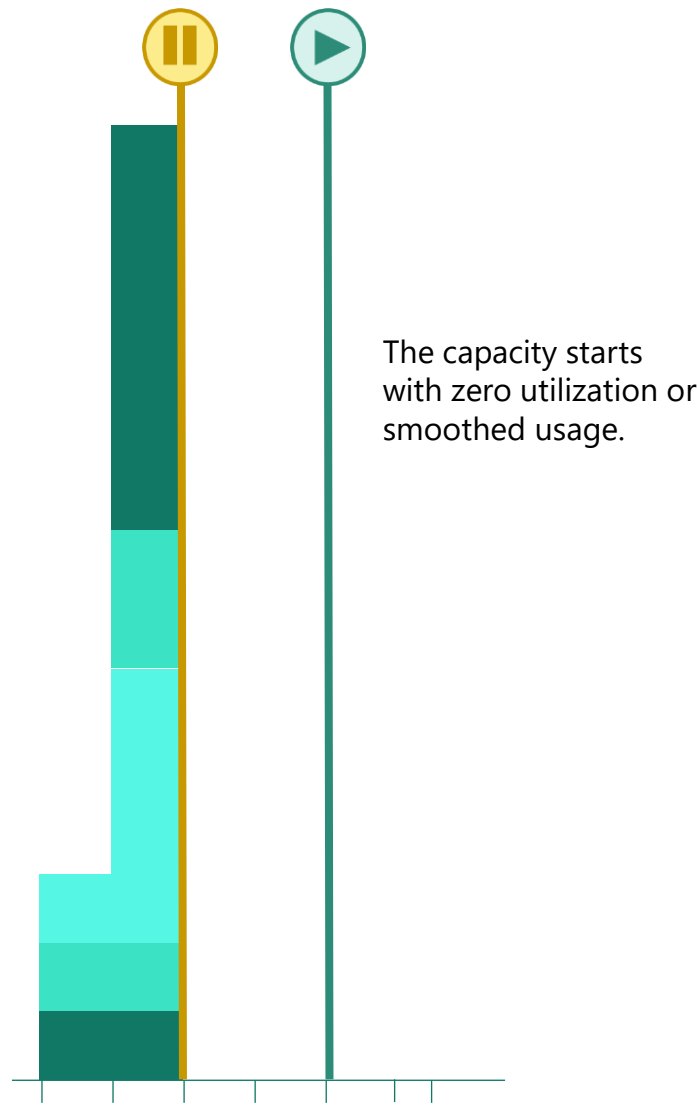
PayGo Price applies to the overage.

A billing event is sent for this
consumed compute.

Smoothed usage is **reconciled**.



Later, it can be **resumed**.



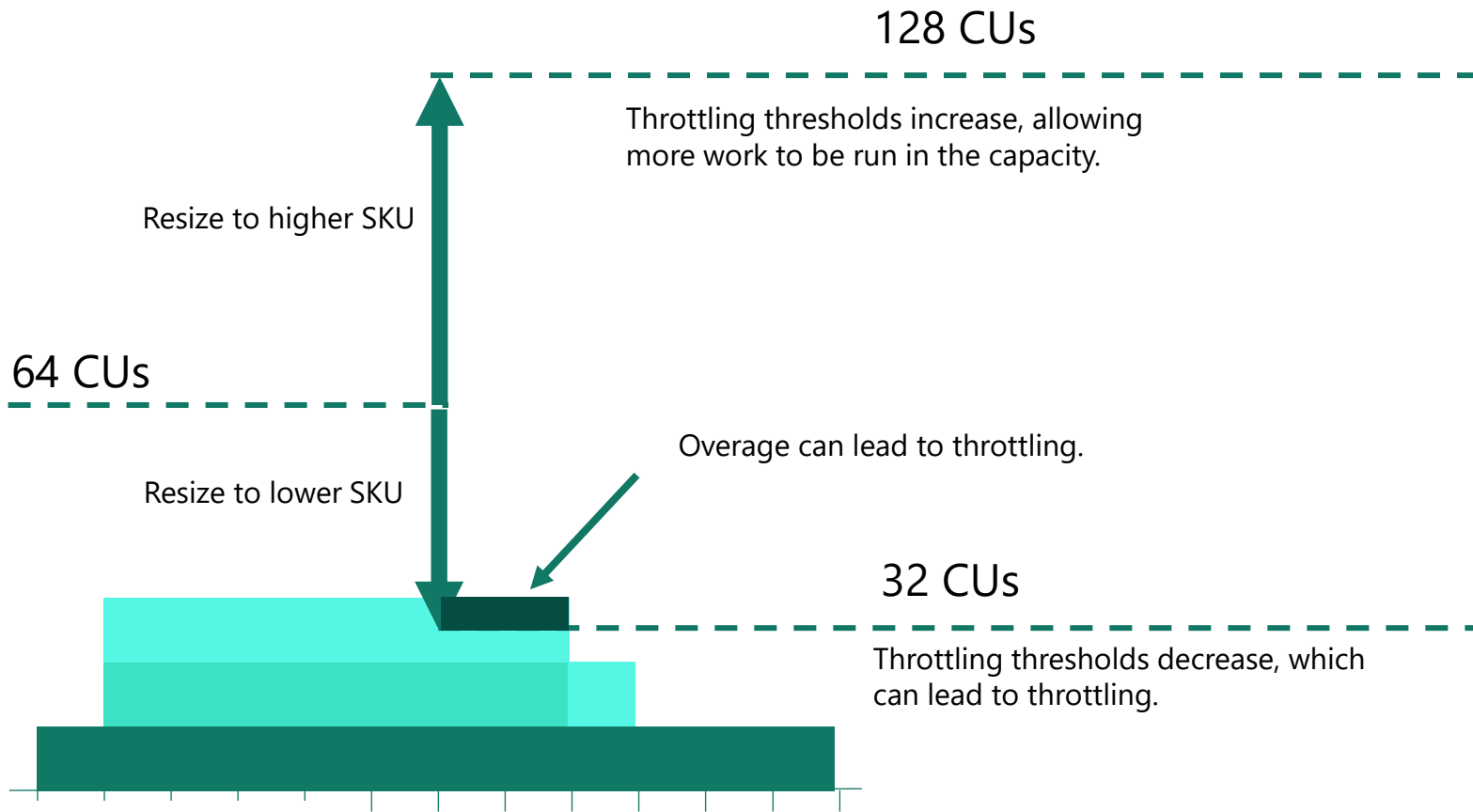
How Capacity Resize works

When a capacity is **resized**...

SKU Change

The allowed CUs per timepoint increase or decrease.

This changes the throttling allowed limits based on the new SKU's CUs and the throttling windows.



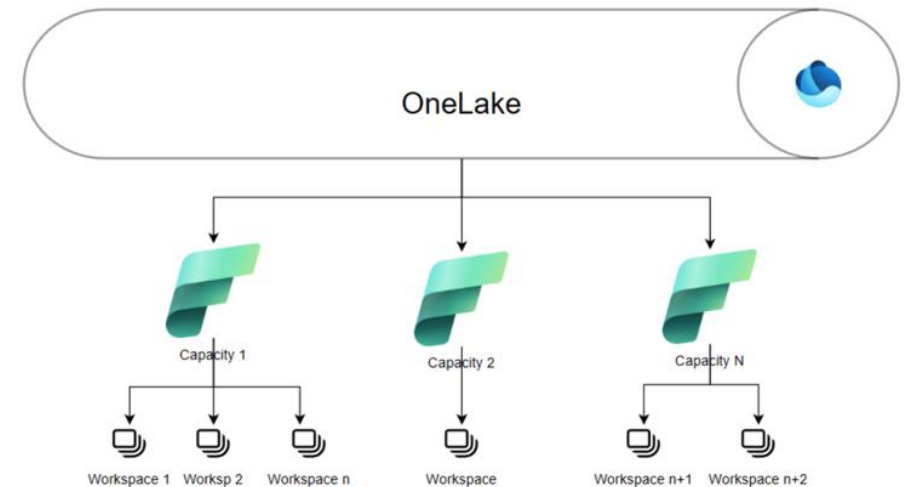
Key Insights

- Sizing up will incur the cost of the new SKU
- Sizing down could lead to more throttling
- Review your Throttling Thresholds before sizing down your SKU.

How do you prevent an overload ?

General cost saving recommendations

- Leverage a **multi-capacity strategy** (isolate, tryout, timeout, etc.)
- Invest in user education, **best practice sharing**, CoE, etc. (proactive optimization)
- Avoid data/report sprawl by **maximizing reuse** of master data and reports (leverage endorsement, discoverability, OneLake shortcuts, etc.)
- Right size your capacities and leverage **pause / resume / resize**, or Reserved Instances for **discounts**. Consider a combo of RI and PAYGO (for predictable surge activity)
- Choose the right tool for the job and stay up to date on Fabric feature releases. e.g. **High concurrency mode for notebooks**
- When relevant, **leave some workspaces in Power BI Pro/PPU**.



How do you prevent an overload ?

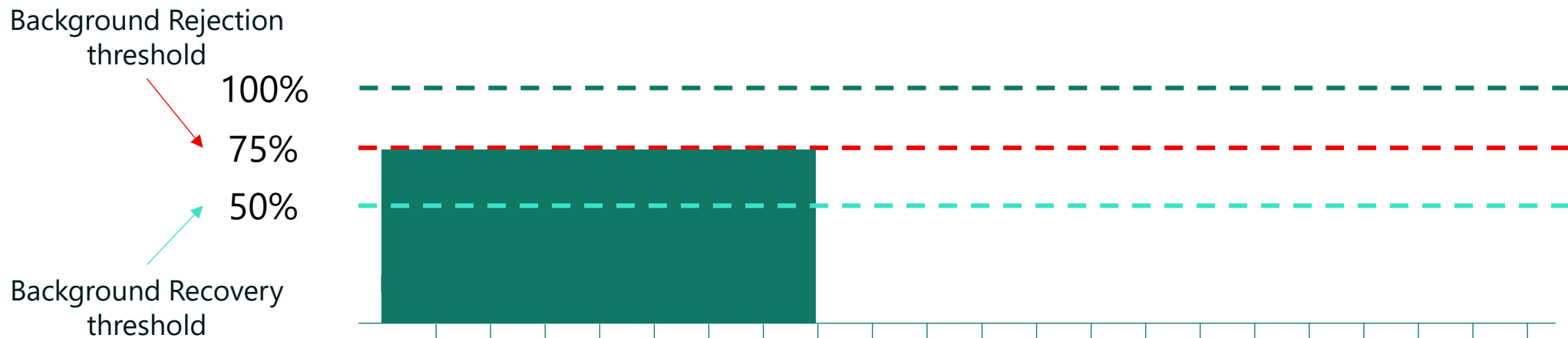
Surge protection

Background rejection threshold

- Throttles background jobs before a full 24-hours of CU is consumed.

Recovery limit

- Keep throttling until the capacity is 'healthy' as defined by the customer
- Helps prevents a capacity from immediately being throttled again



How do you prevent an overload ?

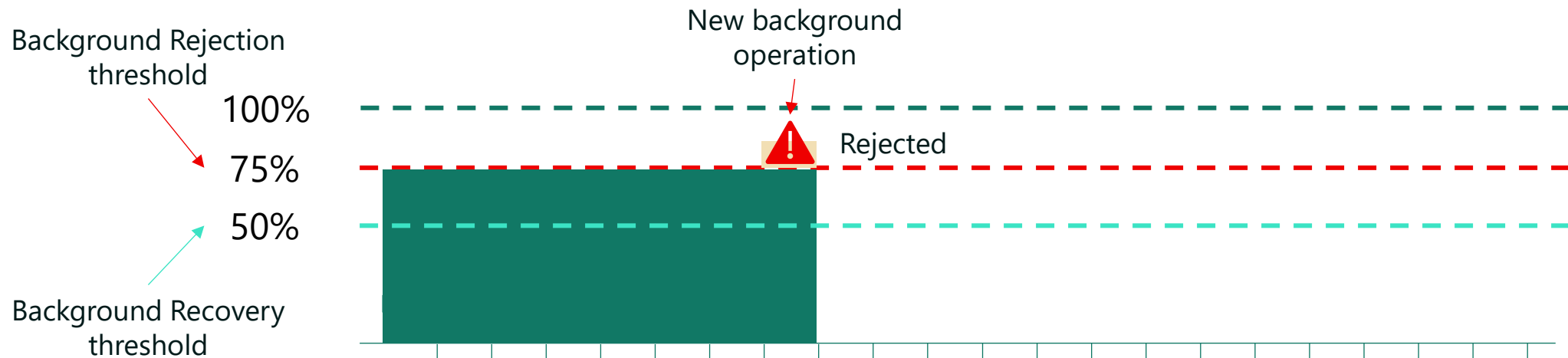
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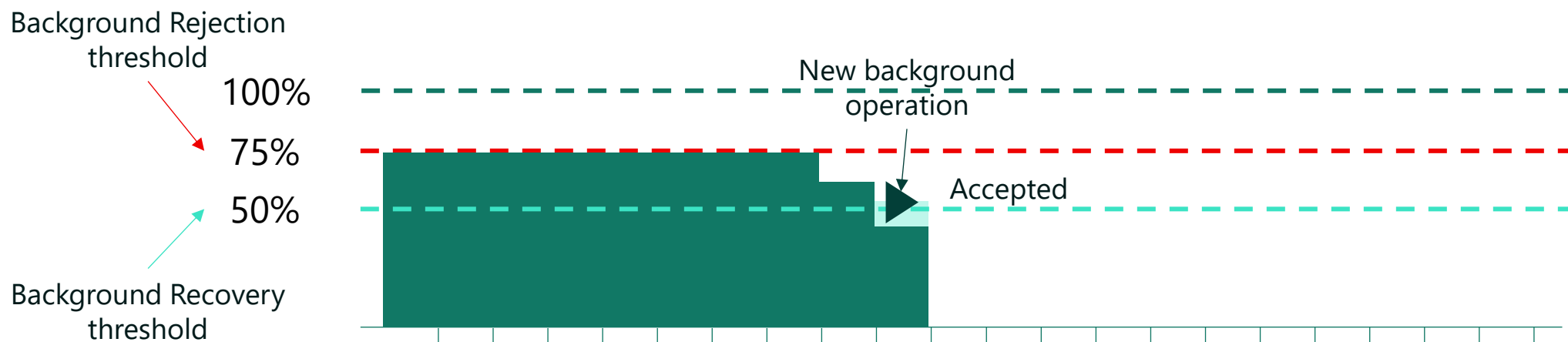
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Fabric Copilot Capacities

[Fabric Copilot capacity](#)
[Introducing Fabric Copilot capacity:](#)
[Democratizing AI usage in Microsoft Fabric](#)

- Capacity that can be enabled for Copilot usage without having items in Capacity backed workspace
- Tenant setting :

Capacities can be designated as Fabric Copilot capacities
Enabled for the entire organization

With this setting on, capacity admins can designate capacities as Fabric Copilot capacities. Copilot capacities are special capacity types that allow your organization to consolidate users' Copilot usage and billing on a single capacity. [Learn More](#)

When users use Copilot features, capacity admins can see the names of the items associated with users' Copilot activity. [Learn More](#)

☒ Enabled

Apply to:

- ☒ The entire organization
- ☐ Specific security groups

Fabric Copilot Capacities

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[Introducing Fabric Copilot capacity:](#)
[Democratizing AI usage in Microsoft Fabric](#)

- Capacity that can be enabled for Copilot usage without having items in Capacity backed workspace
- Tenant setting
- Capacity setting :

Copilot capacity

Disabled for the entire organization

Turn on this setting to designate this capacity as a Fabric Copilot capacity. Copilot capacities are special capacity types that allow your organization to consolidate users' Copilot usage and billing on a single capacity. Copilot capacities may not be available in all regions. [Learn more](#)

Select the users or user groups who can use this capacity for their Copilot usage and billing.

Apply to:

- ☐ The entire organization
- ☒ Specific users or groups



Microsoft Fabric Capacity Metrics

Microsoft Fabric Capacity Metrics

The *Microsoft Fabric Capacity Metrics* app, also known as the *metrics app*, is designed to provide monitoring capabilities for Fabric and Power BI Premium capacities. Use this guide to install the app.

AppSource | Apps for Power BI

< Apps



Microsoft
★★★★★ 2.4 (76)

Get It Now

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Free

Products
Power BI apps

Publisher
Microsoft

Acquire Using
Work or school account

Version
1037

Updated
2/6/2026

Categories
IT & Management Tools

Support
Support
Help

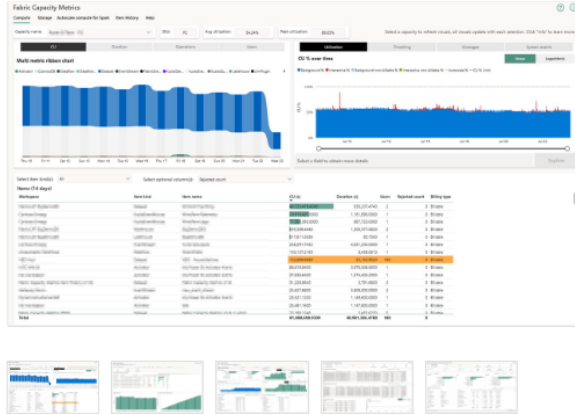
Legal
License Agreement
Privacy Policy

Microsoft Fabric Capacity Metrics

Overview Ratings + reviews

Fabric capacity admins can gain visibility into the resources utilized by their Fabric items

Microsoft Fabric capacity admins can gain visibility into capacity resources. This app enables admins to monitor usage of compute and storage on their capacities. It includes analysis of utilization, processing time and user details for Fabric experiences running on a capacity. Gain visibility into overload causes, peak demand time, resource consumption and easily identify the most demanding or most popular items.



Other apps from Microsoft



Fabric Chargeback

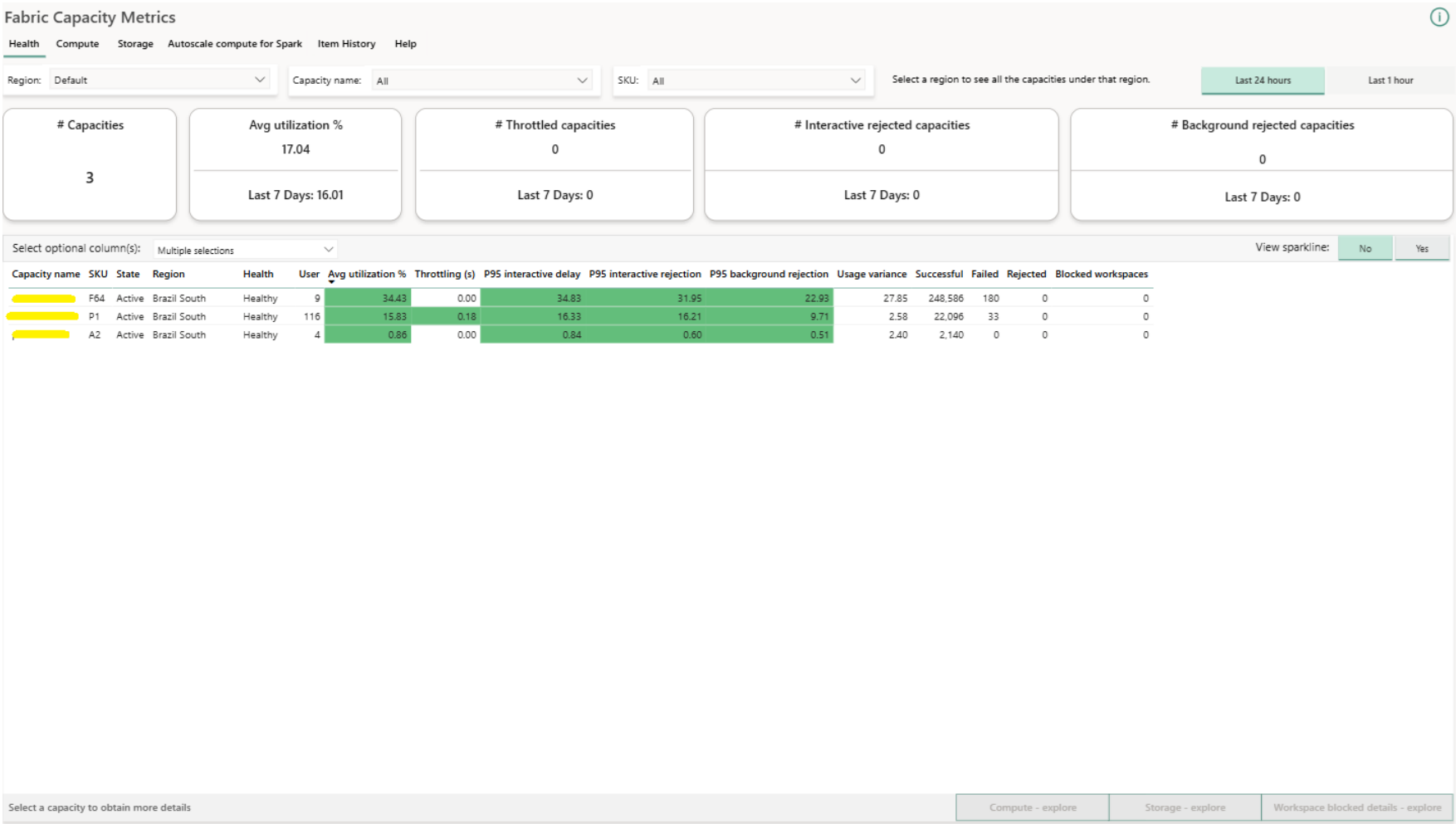
Health Page (Preview)

The **Health** page in the Microsoft Fabric Capacity Metrics app offers a high-level overview of all capacities that a user is admin of.

The **Health** page has two views:

Last 24 hours: Displays metrics aggregated over the past 24 hours.

Last one hour: Displays near real-time insights over the past 60 minutes.

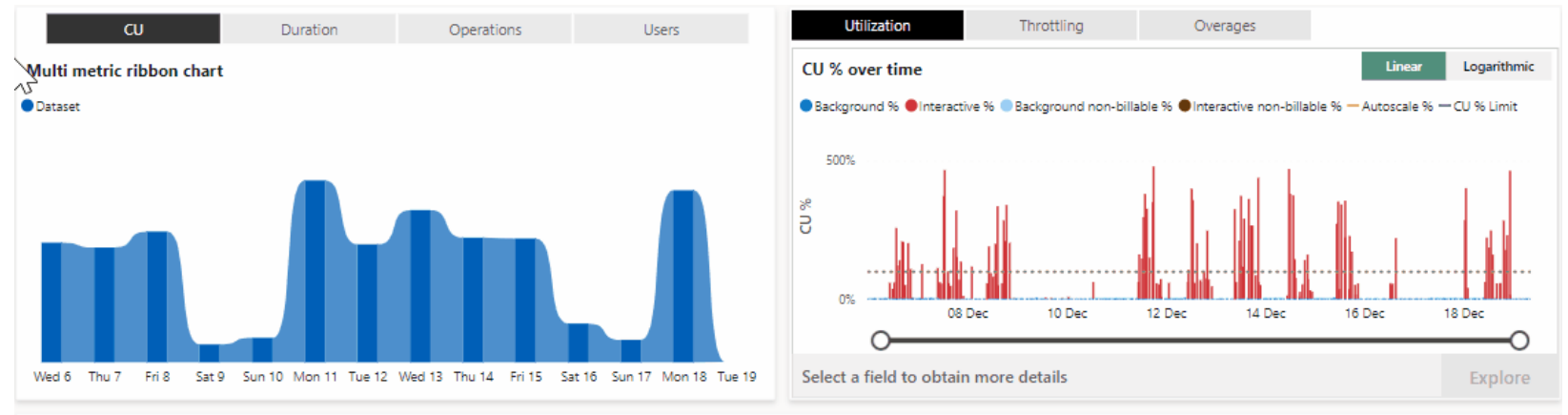


Compute Page

Compute page provides an overview of your capacity's performance, showing data for the last 14 days

Utilization

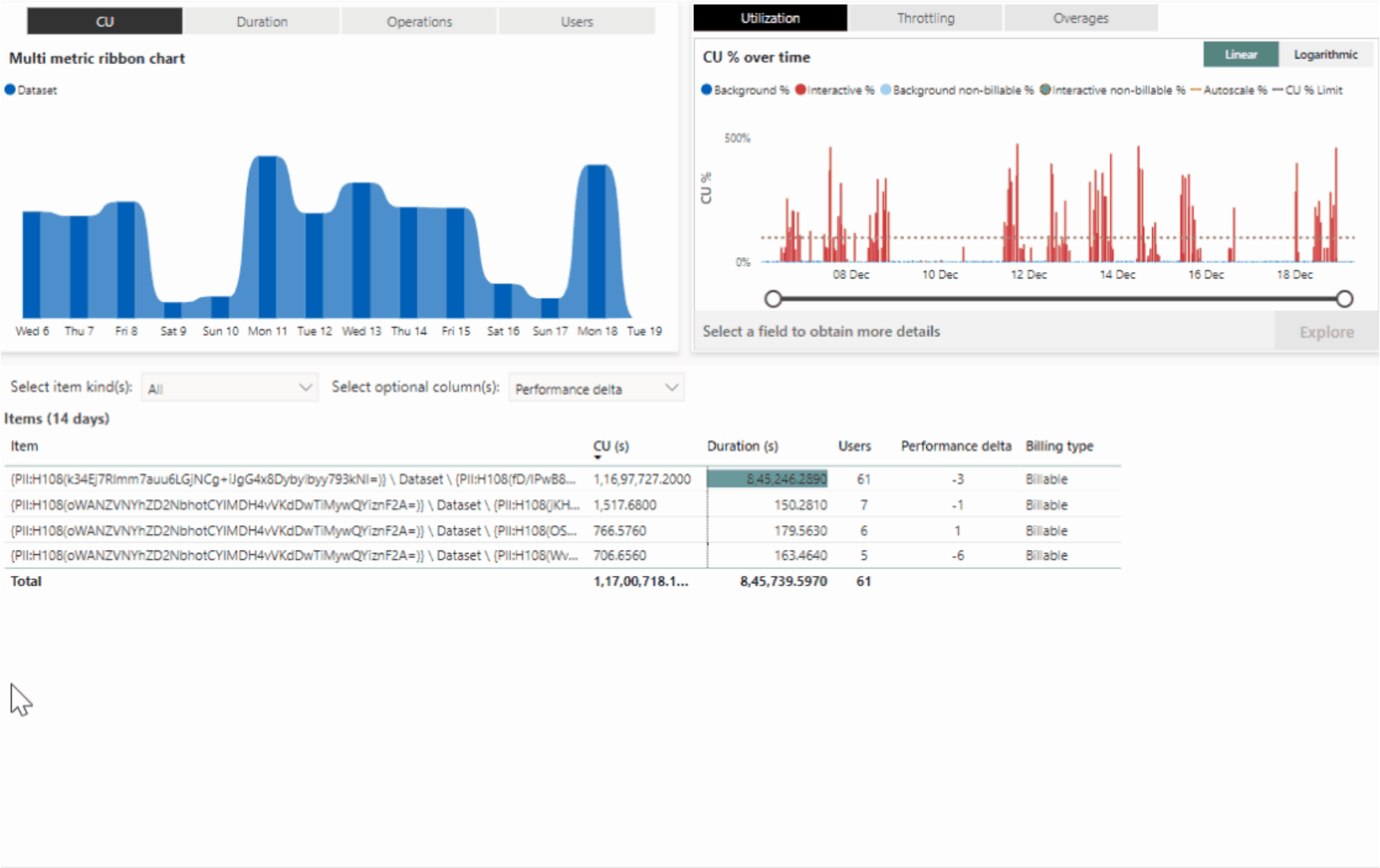
Displays CU usage over time.



Compute Page

Throttling

Displays delay and rejection over time.



Utilization

Throttling

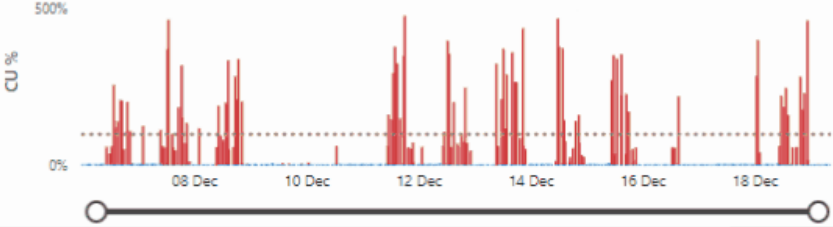
Overages

CU % over time

Linear

Logarithmic

Background % Interactive % Background non-billable % Interactive non-billable % Autoscale % CU % Limit



0% 500%

08 Dec 10 Dec 12 Dec 14 Dec 16 Dec 18 Dec

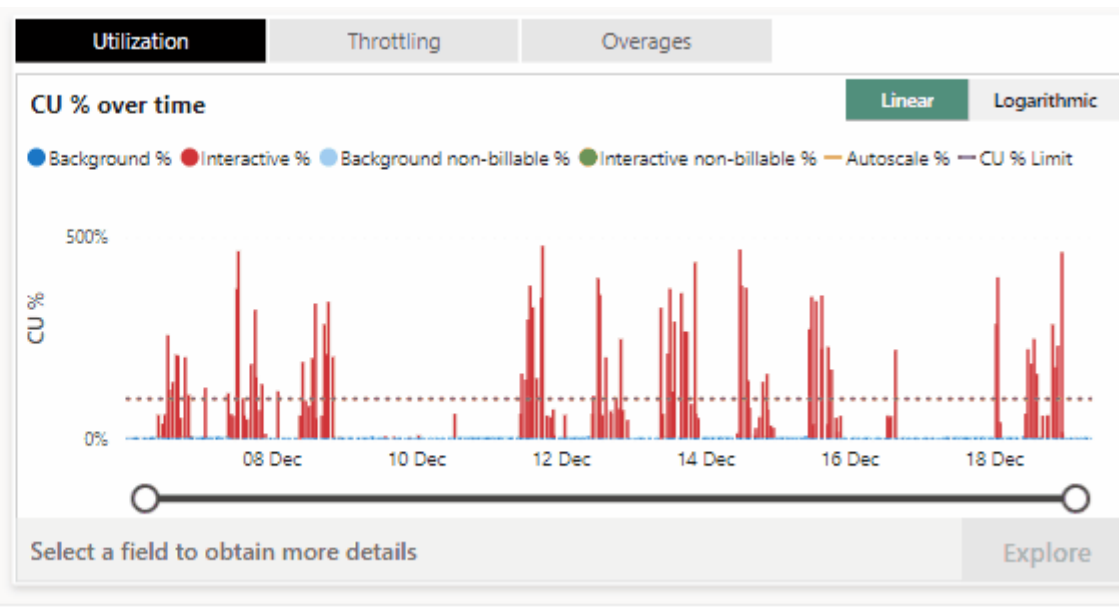
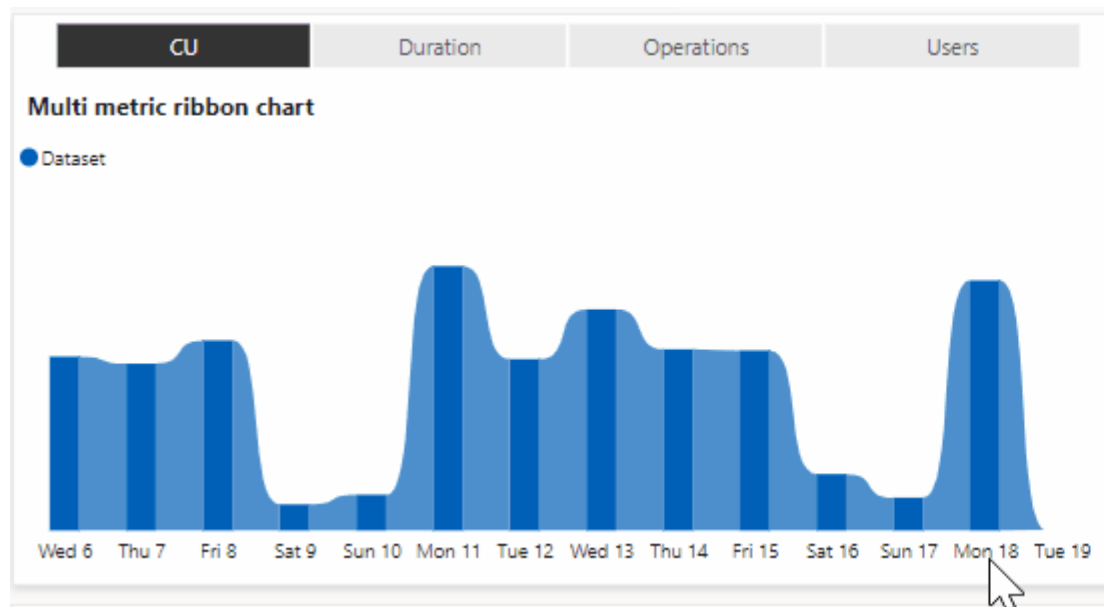
Select a field to obtain more details

Explore

Compute Page

Overages

Displays the *add*, *burndown*, and *cumulative* carryforward over time. Carryforward only takes into account billable operations.



Compute Page

Matrix by item and operation

A matrix table that displays metrics for each item on the capacity. To gain a better understanding of your capacity's performance, you can sort this table according to the parameters listed in this section.

The colors in the table represent your [performance delta](#)

Select item kind(s):

All

Select optional column(s):

Rejected count

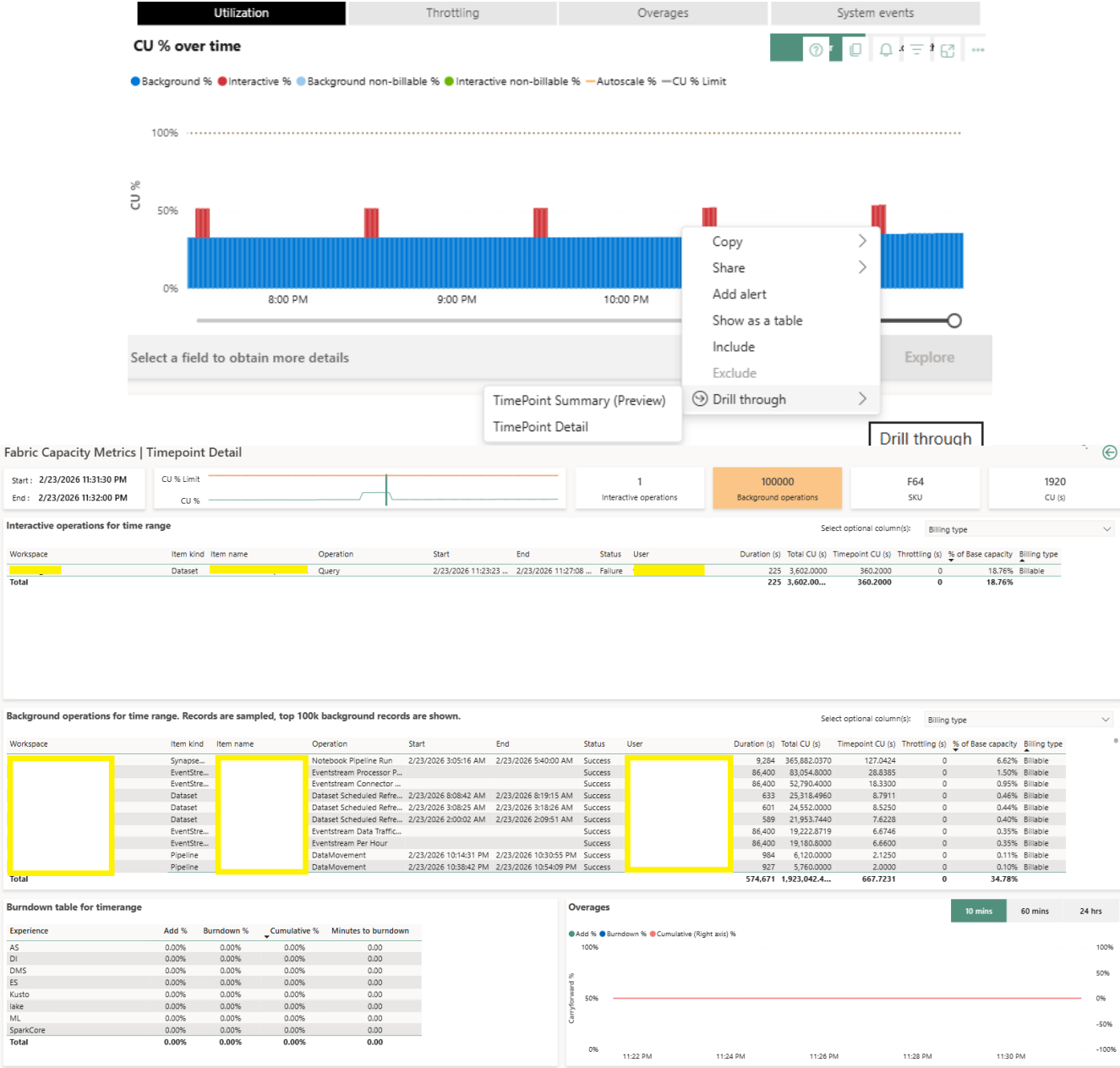
Items (1 day)

Workspace	Item kind	Item name	CU (s)	Duration (s)	Users	Rejected count	Billing type
	Lakehouse		851,060.4056	50.4780	1	0	Billable
	SynapseNotebook		365,882.0370	9,284.8930	1	0	Billable
	KustoEventHouse		191,430.0000	85,080.0000	1	0	Billable
	EventStream		176,198.4730	348,000.0000	1	0	Billable
	Pipeline		164,934.7200	40,700.4180	2	0	Billable
	Dataset		134,633.1040	6,714.0150	8	0	Billable
	Warehouse		56,092.1380	97,470.4200	1	0	Billable
	Dataset		24,712.0000	612.3280	2	0	Billable
	Pipeline		15,537.6000	9,758.6900	1	0	Billable
	Pipeline		14,276.1600	8,196.5340	1	0	Billable
	Warehouse		10,396.9626	1,253.1320	7	0	Billable
	Lakehouse		7,330.3052	14.2410	1	0	Billable
	Lakehouse		7,019.5854	11.5980	1	0	Billable
	Lakehouse		5,704.7613	1,273.7180	2	0	Billable
	SQLDbNative		5,568.0752	14.8990	1	0	Billable
Total			2,080,982.9952	995,604.7600	8	0	

Compute Page

Timepoint Detail

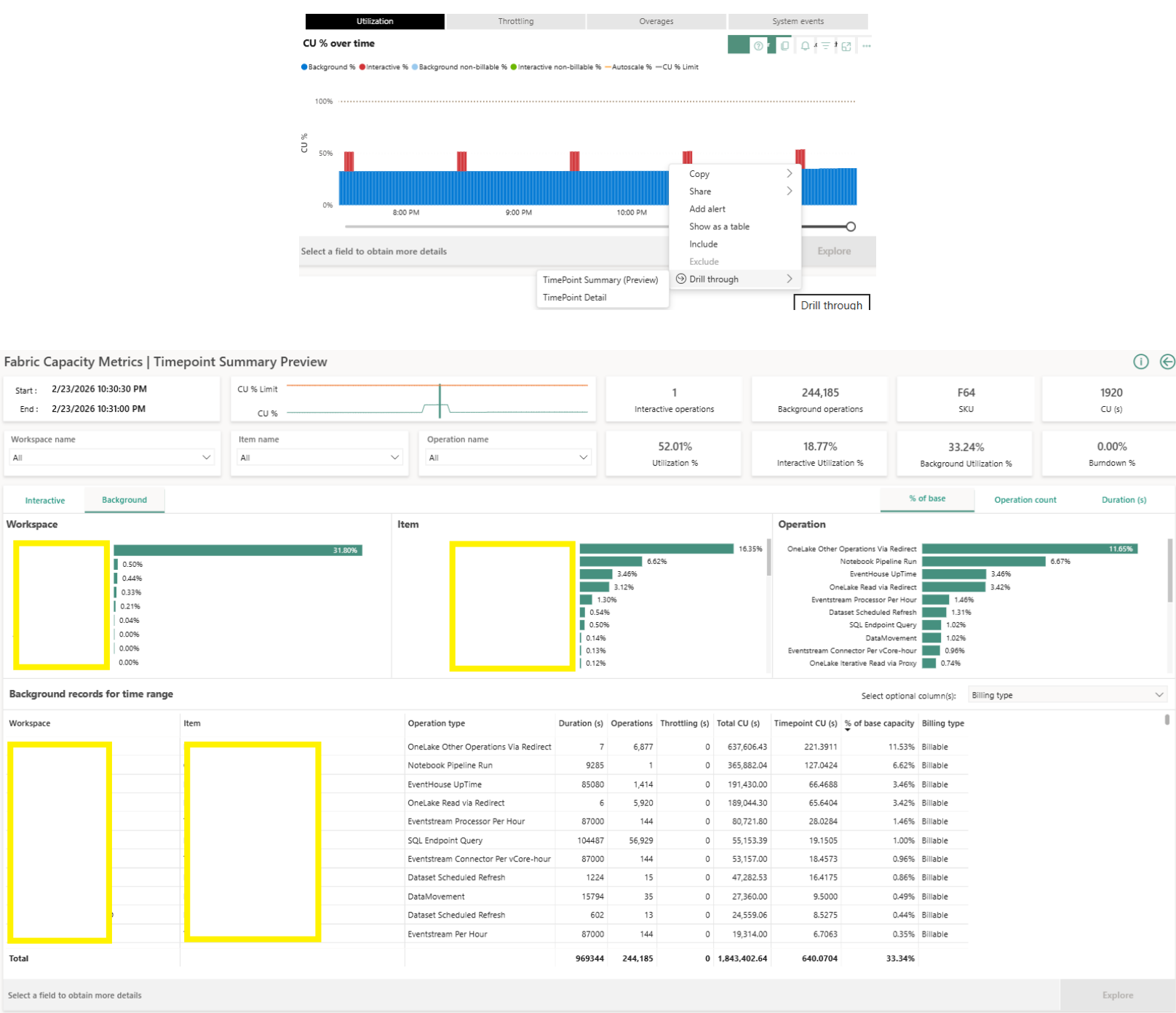
Use this page to understand which interactive and background operations contributed the most to your capacity's usage.



Compute Page

Timepoint Summary (Preview)

This page provides a summary view of **operation types** (not individual operations) in items that contribute to capacity usage in the 30-seconds time window, ranked by their compute impact measured in **Capacity Units (CUs)**. Use it to identify which interactive and background operation types had the highest impact on your capacity.



Compute Page

Timepoint Item Detail (Preview)

This page provides a detailed analysis of the operation type within a specific item used to drill through to this view. If the selected workspace, item, and operation type contain more than 100,000 records for **Interactive** or **Background** operations within the timepoint window, sampling occurs.

Background records for time range

Select optional column(s): Billing type

Workspace	Item	Operation type	Duration (s)	Operations	Throttling (s)	Total CU (s)	Timepoint CU (s)	% of base capacity	Billing type
		Query	2	8	0	22.78	0.0079	0.00%	Billable
		Query	1	2	0	12.26	0.0043	0.00%	Billable
Total			2	10	0	35.04	0.0122	0.00%	

Select a field to obtain more details

Explore

Fabric Capacity Metrics | Timepoint Item Detail

Start : 2/23/2026 10:30:30 PM
End : 2/23/2026 10:31:00 PM

Item name

Dataset
Item kind

Workspace name

Query
Operation

Operation Id
All

User
All

Status
All

18.77%
Interactive utilization %

0.00%
Background utilization %

1
Interactive operations

8
Background operations

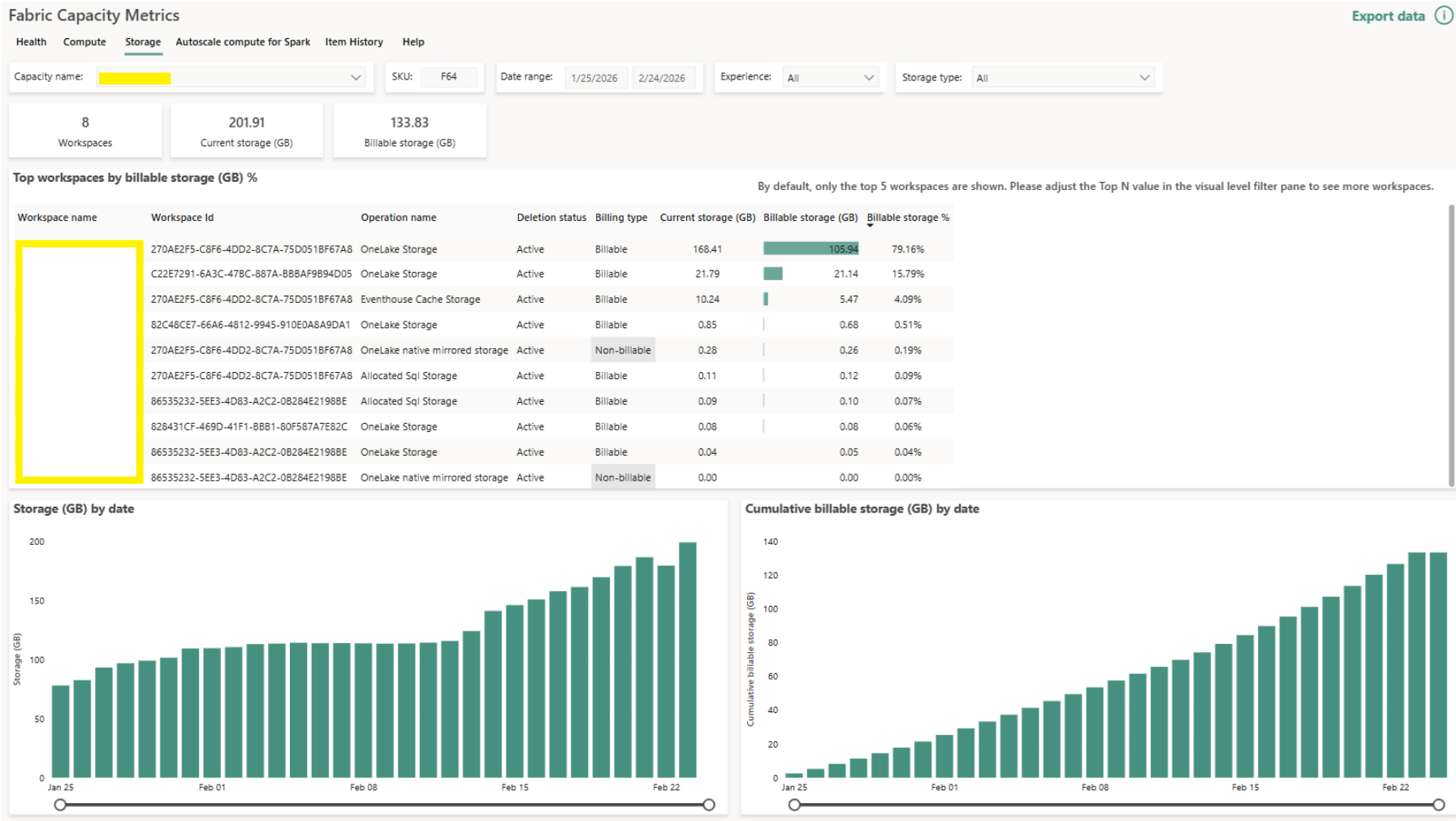
Background operations for time range

Select optional column(s): Billing type

Start	End	Status	User	Duration (s)	Throttling (s)	Total CU (s)	Timepoint CU (s)	% of base capacity	Operations	Billing type
2/23/2026 8:19:16 AM	2/23/2026 8:19:16 AM	Success		0	0	11.76	0.00	0.00%	3	Billable
2/23/2026 2:09:53 AM	2/23/2026 2:09:53 AM	Success		0	0	10.77	0.00	0.00%	4	Billable
2/23/2026 3:10:10 AM	2/23/2026 3:10:10 AM	Success		0	0	0.26	0.00	0.00%	1	Billable
Total				0	0	22.78	0.01	0.00%	8	

Storage Page

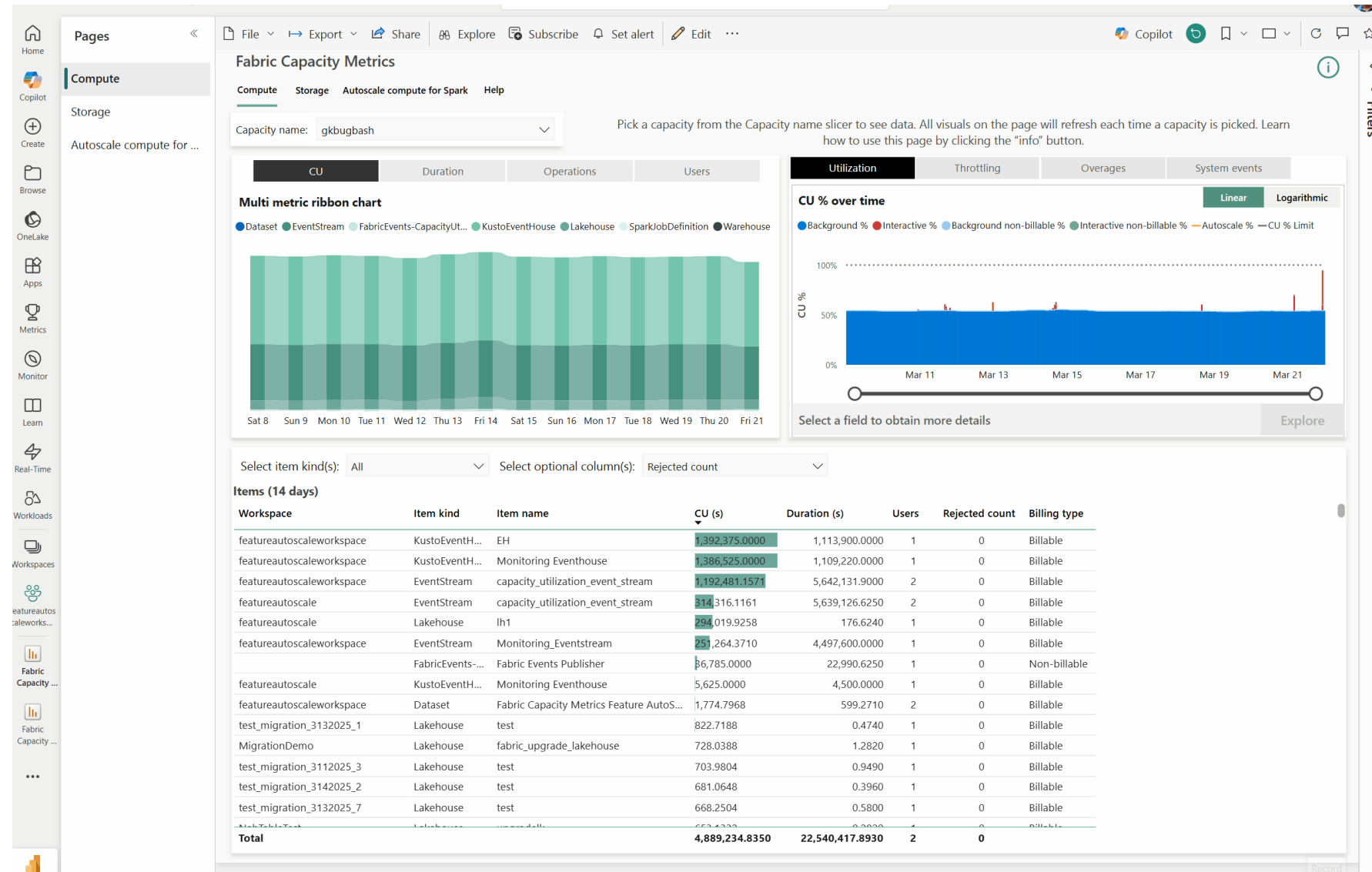
The Microsoft Fabric Capacity Metrics app's storage page provides capacity storage information, showing data for the last 30 days.



Autoscale compute for spark Page

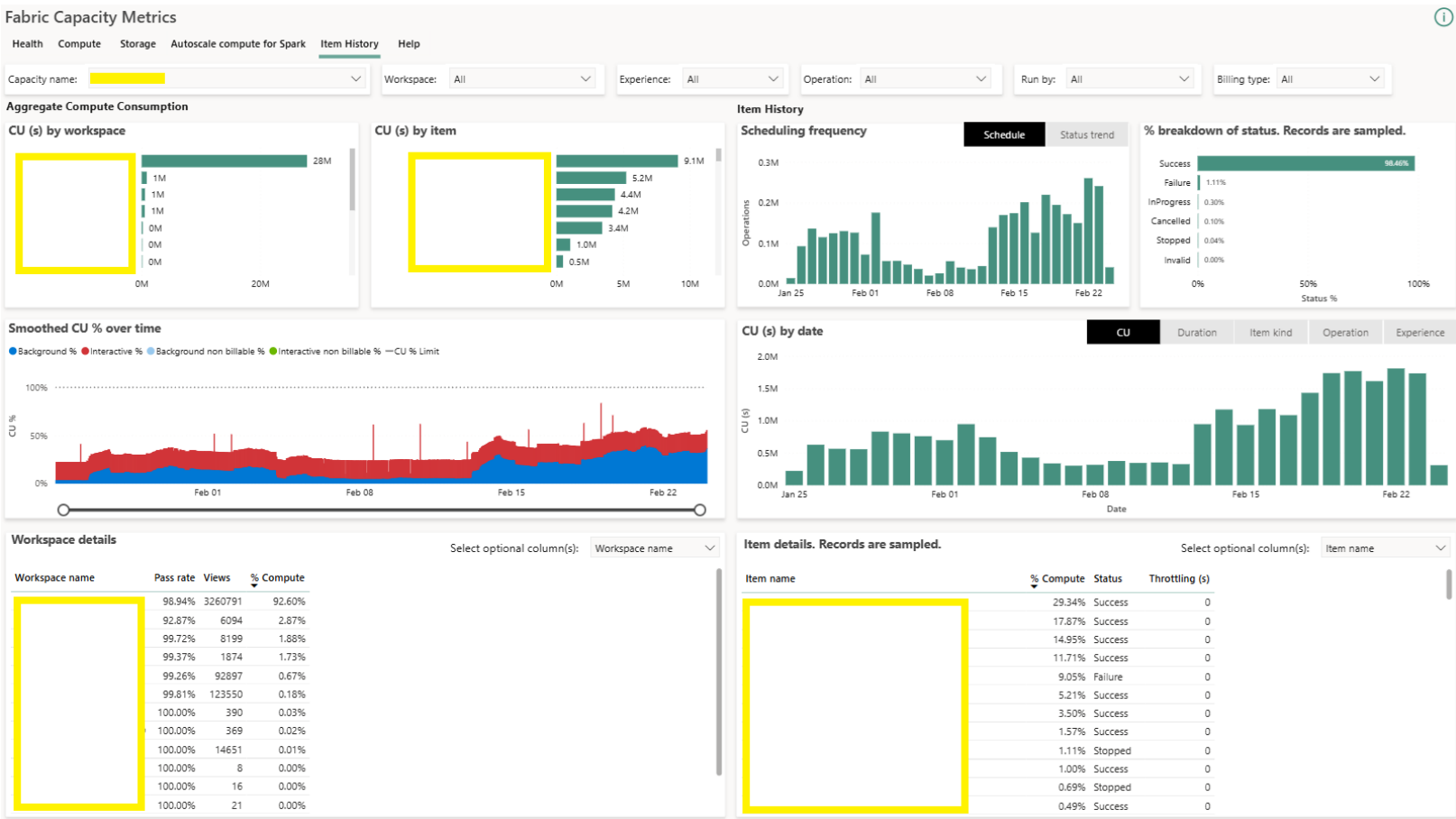
Dedicated Autoscale Billing page that will allow users to:

- Track Spark job usage across workspaces.
- Monitor overall usage trends at different points in time.
- Drill down into specific operations to analyze job run duration and resource allocation.



Item history (Preview)

The **Item history** page provides compute usage analysis through slicers and dynamic visuals for the last 30 days. It presents high-level consumption trends alongside detailed item-level metrics, enabling users to explore and monitor operational and billing data effectively.



Capacity Mitigations



How to remedy an overload ?



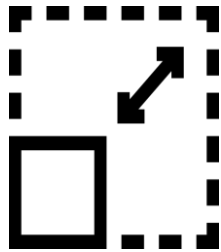
Notifications



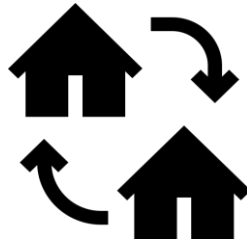
Artifact
Management



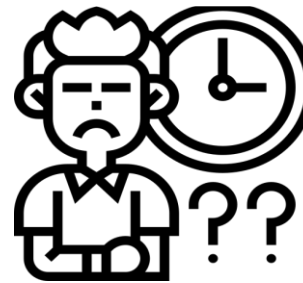
Endorsement
Process



Autoscale



Capacity Re-
Allocation



Wait...

1/6 Notification

- Alerts related to a Fabric capability can be viewed through the tenant Fabric Admin portal.

- Each capability may have its own notification settings configured separately.

Admin portal

Tenant settings New

Users

Premium Per User

Audit logs

Domains New

Workloads

Tags New

Capacity settings

Refresh summary

Embed Codes

Organizational visuals

Organizational themes (preview)

Azure connections

Workspaces

Custom branding

Fabric identities

Featured content

Help + support

Notifications

Send notifications when

☒ You're using % of your available capacity

☒ You've exceeded your available capacity and might experience slowdowns

Send notifications to

☒ Capacity admins

☐ These contacts:

Apply

Discard

Contributor permissions

Disabled for the entire organization

Admin permissions

Power BI workloads

Preferred capacity for My workspace

Data Engineering/Science Settings

Autoscale Billing for Apache Spark

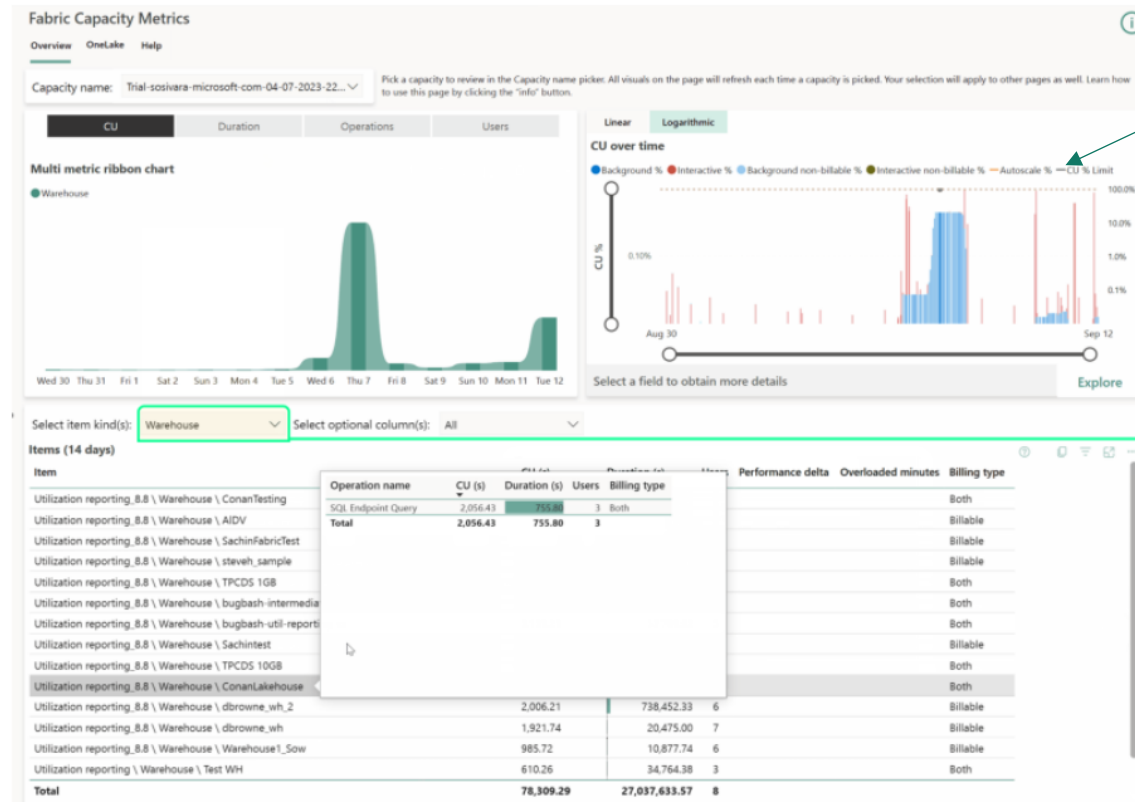
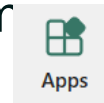
Workspaces assigned to this capacity

2/6 Artifact management

- Microsoft offers a standard application called the Fabric Capacity Metrics App to identify the largest capacity users.

- It is available for download from the “Apps” section.

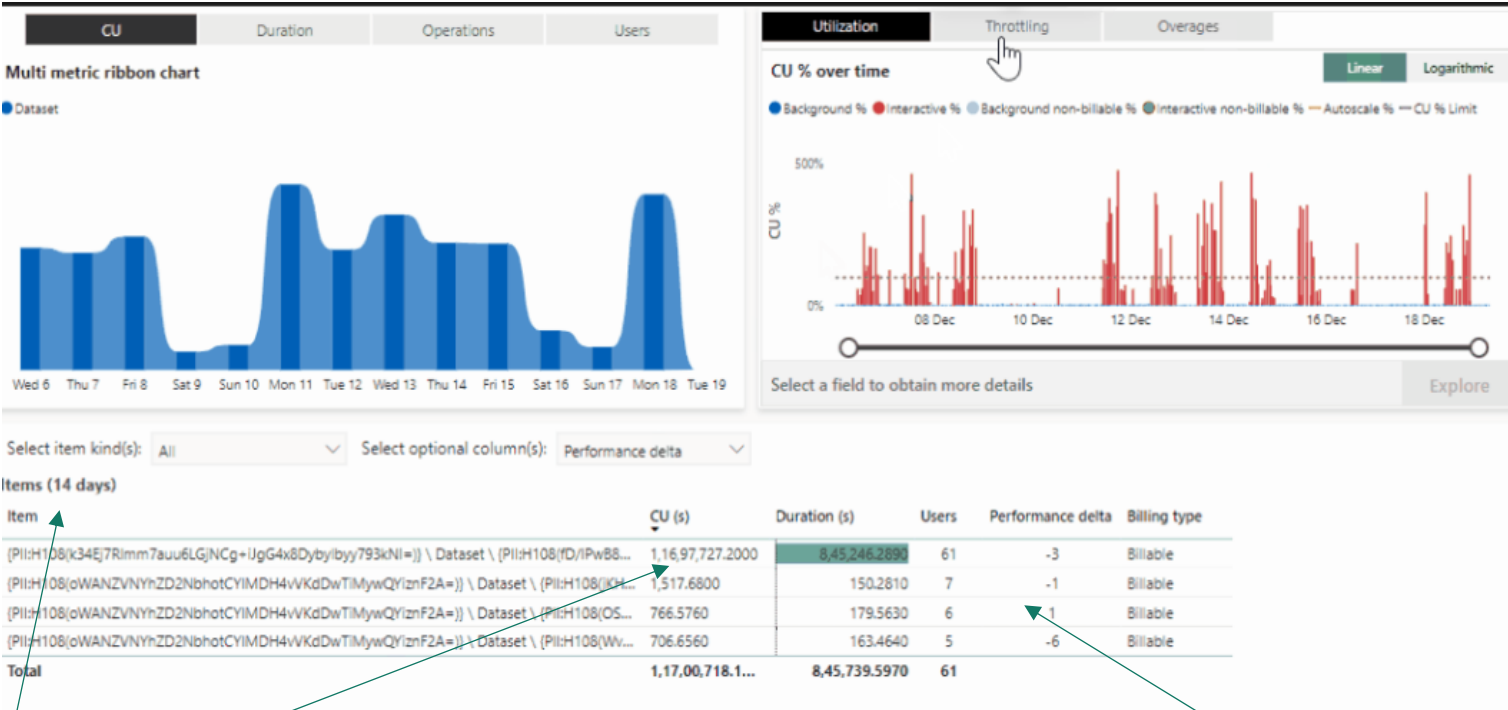
- Ensure you link the app to a “pro” license to prevent any throttling effects.



Shows how much capacity your Fabric content is consuming against the limit you pay for

2/6 Artifact management

- CU = Consumption Unit
- F64 Capacity provides 64 CUs for every second (i.e 5.5 m CUs per day)



In 14 Days it consumed more than 11 millions CUs

Shows consumption at artefact level

2/6 Artifact management



Reactive

- Alert
- Reviewing peaks
- Identifying problematic artefacts
- Taking immediate action



Proactive

- Not triggered by an alert
- Reviewing artefacts with large CU use over 14 days
- Discuss with owners

2/6 Checklist procedure for each situation

100% +

- Crisis mode
- Immediately reduce consumption
- All options open – e.g stop On prem data gateways, cancel refreshes, remove access to problematic content, Increase Capacity size

80 – 99%

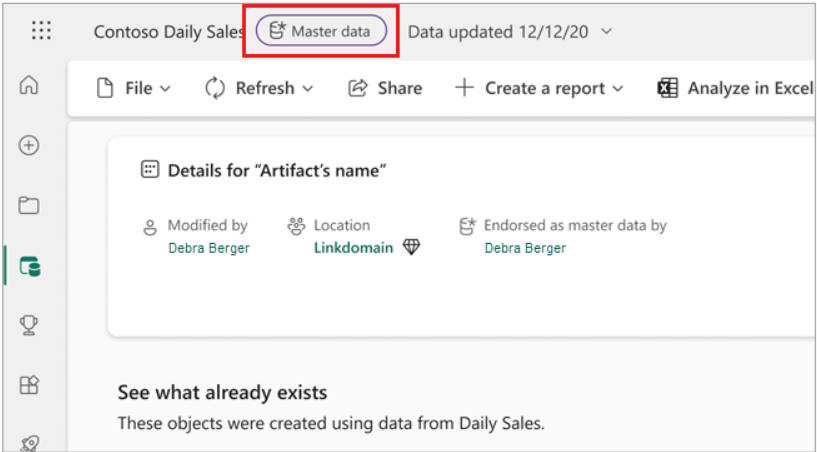
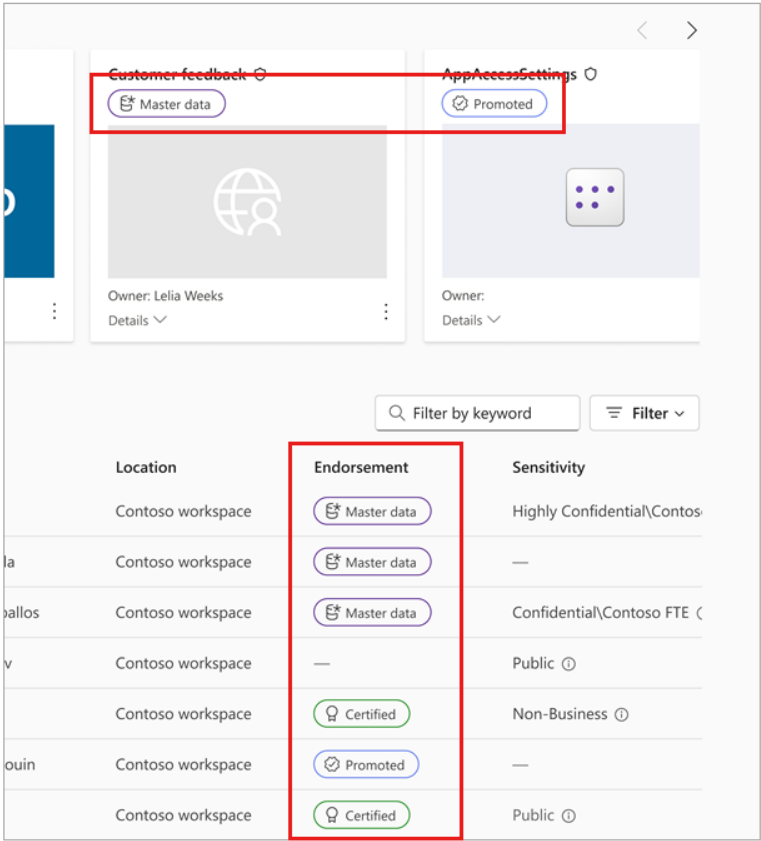
- Crisis avoidance
- Identify problematic artefacts
- Understand what happened and react
- Take reasonable steps to reduce consumption

60 – 79%

- Review situation and decide if action require

3/6 Endorsement

- Endorsement is a Fabric and Power BI feature that makes it easier for users in an organization to find high-quality, trust-worthy content and data. Endorsed items are clearly labeled in the UI with a badge, and in some lists they are given precedence and are listed first.



4/6 Moving to Shared Capacity

Power BI Premium feature list

Expand table

Feature	Description
Asynchronous refresh	Perform asynchronous data-refresh operations
Automatic aggregations	Optimize DirectQuery datasets
Autoscale	Automatically add compute capability when your capacity is overloaded Available for Premium only, excluding EM SKUs
Backup and restore	Backup and restore data using XMLA endpoints
Bring your own key (BYOK)	Use your own keys to encrypt data Available for Premium and Embedded
Dataflows	- Perform in-storage computations - Optimize the use of dataflows - Use incremental refresh with dataflows - Reference other dataflows
Datamarts	Self-service solution enabling users to store and explore data that's loaded in a fully managed database
Deployment pipelines	Manage the lifecycle of your Power BI content
Direct Lake (preview)	Connect directly to your data lake without having to import its data
DirectQuery with dataflows	Connect directly to your dataflow without having to import its data
Hybrid tables	Incremental refresh augmented with real-time data
Insights (preview)	Explore and find insights such as anomalies and trends in your reports
Model size limit	Available memory is set to: <i>Premium</i> - The limit of memory footprint of a single Power BI dataset; see the column <i>Max memory per dataset</i> in the Capacities and SKUs table <i>Premium Per User (PPU)</i> - See Considerations and limitations <i>Embedded</i> - See the column <i>Max memory per dataset</i> in the SKU memory and computing power

- When reviewing an artifact, you might find that it does not need a Capacity
- For this :
 - No Premium features
 - No Fabric artefacts
 - Audience has PowerBI Pro or PPU License

4/6 Multiple Capacities (Example)

Capacity	Content size	Nature of content	Environment	Typical Utilisation
Self Service (P1)	100 workspaces 1500 Reports	Dataflows, models and report, Critical HR content	Dev, Test, UAT and production	80%
Enterprise Content (P2)	220 workspaces 2200 reports	Dataflows, models and reports	Dev,Test UAT and production	30-45%

As an Admin, analyze your utilization against administration and business requirements. You may need more capacity or more a shift in management strategy!

5/6 AutoScale



Fabric Scale

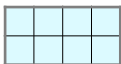


Power BI Autoscale ?



Power BI

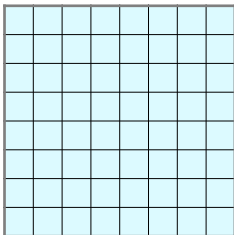
F8 Capacity



8 CU's



F64 Capacity



64 CU's

Autoscale settings

Autoscale is an optional add-on that leverages your Azure subscription to automate your response to unplanned overage spikes in Power BI Premium. Once you come up against your capacity limit, we'll add one v-core per 24-hour period to prevent overload slow-downs. Set your own cost limits to prevent overruns on v-core scaling or total Azure subscription charges.

[Learn more about pricing](#)

☒ Enable Autoscale

Select an Azure subscription to use for billing

Subscription

Azure Sponsorship -

Resource group

powerBIPremiumAutoscaleCores

Set an Autoscale maximum

To control costs, limit the number of additional v-cores your capacity can scale up to.

Autoscale max

8

+

-

v-cores

Save

Cancel

Key Takeaways

Monitor Capacity Proactively

Drive best practices (Endorsement)

Checklist of actions to do when X situation

Multiple capacities > 1 large capacity

Invest in user education (proactive optimization)

Choose the right tool for the job

stay up to day on fabric features (High concurrency mode for notebook)